



FOREWORD



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It is with great pleasure that I present the “ICAR Kharif Agro-Advisories for Farmers 2025”- a timely and comprehensive guide developed to empower India’s farming community with science-based, region-specific advisories for the upcoming Kharif season. This publication marks a significant step toward strengthening climate-resilient agriculture and enhancing farm productivity across diverse agro-climatic zones.

Agriculture in India stands at a critical juncture. While it continues to serve as the backbone of rural livelihoods, it faces unprecedented challenges due to climate variability, resource degradation, pest dynamics, and market volatility. In this context, the role of timely and location-specific agro-advisories has become more crucial than ever. These advisories act as a bridge between cutting-edge agricultural research and its practical application in farmers’ fields.

This publication offers practical guidance on region-based suitable varieties, nutrient management, pest and disease control, and harvesting practices tailored for major Kharif crops such as rice, maize, pulses, oilseeds, and cotton. It is the result of collaborative efforts by ICAR Institutes, Agricultural Universities, Krishi Vigyan Kendras (KVKs), and Subject Matter Experts working at the grassroots level.

At ICAR, we remain committed to the vision of a knowledge-driven, inclusive, and sustainable agricultural system. The advisories compiled herein reflect our continuous efforts to translate scientific knowledge from laboratories into actionable insights for every farmer. I urge extension officers, KVK teams, farmer producer organizations (FPOs), and progressive farmers to widely disseminate and implement the recommendations provided in this book.

I extend my heartfelt appreciation to all scientists, experts, and field professionals who have contributed to the preparation of this publication. I am confident that this agro-advisory compilation will serve as a valuable resource for farmers during upcoming Kharif season, enabling them to make informed decisions that lead to improved productivity, profitability, and sustainability.

Let us move forward together to build a resilient, prosperous, and empowered farming community #Viksit Bharat 2047.

Jai Kisan, Jai Vigyan!

(M.L. Jat)

INTRODUCTION

Agriculture remains the cornerstone of India's economy, supporting nearly half of the population and providing employment, nutrition, and rural development. The Kharif season, spanning from June to October, is particularly vital due to its reliance on the south-west monsoon. Approximately 55% of India's net sown area is rainfed, making the timely and equitable distribution of monsoon rains crucial for agricultural success.

In 2025, the India Meteorological Department (IMD) forecasts an above-normal monsoon season, with cumulative rainfall expected at 105% of the Long Period Average (LPA) of 87 cm. The monsoon's onset is anticipated to be slightly delayed in southern India but should reach Maharashtra by May 27, facilitating timely sowing preparations. The IMD's forecast of an above-normal monsoon offers a promising outlook but still carries risks of intra-seasonal dry spells and localized flooding. The Indian government's initiatives, such as the "Panchayat Mausam Seva," aim to deliver hyper-local weather forecasts at the panchayat level, ensuring that farmers receive accurate and timely information.

The Kharif season witnesses the cultivation of a diverse array of crops across India's agro-climatic zones, including rice, maize, cotton, soybean, groundnut, bajra, jowar, tur, urad, and moong. Despite advancements in agricultural technology, the sector remains vulnerable to climatic fluctuations such as delayed rains, uneven distribution, and extreme weather events. These challenges can adversely affect crop yields, farm incomes, and rural livelihoods.

To mitigate these risks, it is imperative to provide farmers with location-specific, research-backed guidance. Agro-advisories play a pivotal role by offering timely, region-specific, science-based information on weather patterns, crop management, pest control, harvesting, and storage. These advisories integrate seasonal outlooks with crop calendars and state-wise sowing windows, enabling farmers to select appropriate crop varieties and sowing dates, optimize resource use, and prevent yield losses.

This advisory serves as a crucial tool for agricultural officers, Farmer Producer Organizations (FPOs), and NGOs involved in rural development. By bridging the information gap between research and practice, it empowers farmers with the knowledge needed to make informed decisions, thereby contributing to the vision of a self-reliant, prosperous, and inclusive India @ VIKSIT BHARAT 2047.

CEREAL CROPS

Paddy

Varieties

North Coastal Zone: Pushkala, Srisatya, Cotton Dora Sannalu, Tarangini, Nellore Masoori, Nellore Dhanyarashi, NLR Rice 3238.

Southern Zone: Sweta, Bharani, Somasila, Nellore Masoori, Nellore Dhanyarashi, Nellore Sugandha, NLR 3238, Cotton Dora Sannalu, Tarangini, Chandra.

HAT Zone: Cotton Dora Sannalu, Tarangini, Chandra, Pushkala, Srisatya, NLR 3238.

Krishna Zone: Swarna, Indra, Amara, Krishnaveni, Varam, MTU Rice 1318, MTU 1232, Panduranga, MCM Rice 103, Samba Masoori, Akshaya, Bhavapuri Sannalu, Bapatla Masoori, Nellore Siri, Teja, Bhavathi, Sasya, BPT Rice 2841 & 2846, Maruteru Samba, Maruteru Masoori, MTU Rice 1271 & 1278, Nellore Sona, Nellore Siri, NLR 3238.

Late Nursery Transplanting: Swarna, Indra, Amara, Maruteru Masoori, MTU Rice 1318, Bapatla Masoori, Bhavapuri Sannalu, Teja, Panduranga, MCM Rice 103. BPH (Brown Plant Hopper) Infestation-Prone Areas: Krishnaveni, Indra, Amara, Maruteru Samba & Masoori, MTU 1271, Bhavapuri Sannalu, Teja, Sasya, BPT Rice 2846, Nellore Siri.

Flood-Prone Areas: Amara, Indra, Bhima, MTU Rice 1232 & 1318. *Saline Soils:* Indra, Panduranga, MCM Rice 103.

Late Planting: Cotton Dora Sannalu, Tarangini, MTU 1293, Nellore Sona, Nellore Masoori, Nellore Siri, NLR 3238.

Drought-Prone Areas: Cotton Dora Sannalu, Vijetha, Tarangini, Chandar, MTU 1293, NLR 3238.

Godavari Zone: Swarna, Indra, Amara, Bhima, Pushyami, Ksheera, Sravani, Maruteru Samba & Masoori, MTU 1212, 1280, 1281, 1293, 1310, 1321, 1318, 1232,

1271, 1275, Samba Masoori, Teja, Bhavathi, Bapatla Masoori, BPT 2841, MCM Rice 103.

Late Nursery Transplanting: Swarna, Indra, Amara, MTU 1318, Maruteru Masoori, Srikakulam Sannalu.

BPH-Prone: Krishnaveni, Indra, Amara, Pushyami, Maruteru Samba, MTU 1271, BPT 2846.

Flood-Prone Areas: Amara, Indra, Bhima, MTU 1232, 1318.

Saline Soils: Indra, Panduranga, MCM Rice 103.

Late Planting: Cotton Dora Sannalu, Tarangini, MTU 1293, Nellore Siri, NLR 3238.

Drought-Prone Areas: Cotton Dora Sannalu, Vijetha, Tarangini, Chandra, Maruteru Samba, MTU 1293, NLR 3238.

North Coastal Region: Swarna, Indra, Amara, Bhima, Pushyami, Ksheera, Sravani, Maruteru Samba, Masoori, MTU 1310, 1318, 1232, 1271, 1275, Samba Masoori, Teja, Bhavathi, Bapatla Masoori, BPT 2841, 2846, MCM Rice 103.

Late Nursery Planting: Swarna, MTU 1318, Maruteru Masoori, Srikakulam Sannalu.

BPH-Prone Areas: Vijetha, Amara, Indra, Sujatha, Pushyami, Maruteru Samba & Masoori, MTU 1271, Bapatla Masoori, BPT 2646, Nellore Siri.

Flood-Prone Areas: Srikakulam Sannalu, Indra, Bapatla Masoori, Bhima, Amara, Ksheera, MTU 1318, 1232.

Saline Soils: Indra, Panduranga, MCM Rice 103. *Late Planting:* Vasundhara, Cotton Dora Sannalu, Tarangini, Sridhruti, Chandra, MTU 1293, NLR 3238.

Drought-Prone Areas: Sona Masoori, Swarna, Cotton Dora Sannalu, Vijetha, Chandra, Tarangini, Sujatha, Maruteru Samba, MTU 1293.

Southern Region: Nellore Sona, Nellore Masoori, Nellore Dhanyarashi, Nellore Siri, Somasila, Bharani, Sweta, NLR 3238, Samba Masoori, Bapatla Masoori, Teja, Bhavathi, Sasya, BPT Rice-2841, BPT Rice-2846, Varam, Maruteru Samba & Maruteru Masoori, MTU Rice-1232, MTU Rice-1271, MCM Rice-103, Nandyala Sona.

Late Nursery Planting: Parthiva, Bapatla Masoori, Teja, Bhavapuri Sannalu, Maruteru Masoori.

BPH-Prone Areas: Maruteru Samba & Maruteru Masoori, MTU 1271, Bapatla Masoori, BPT 2846, Parthiva.

Saline Soils: Nellore Masoori, Panduranga, MCM Rice 103.

Late Planting: MTU 1293, NLR 3238, Nellore Masoori, Cotton Dora Sannalu, Chandara.

Scarce Rainfall Zone: Nellore Sona, Masoori, Siri, NLR 3238, Samba Masoori, BPT 2841, 2846, Varam, Maruteru Samba & Maruteru Masoori, MTU 1271, MCM Rice 103, Nandyala Sona.

Late Nursery Planting: Bapatla Masoori, Bhavapuri Sannalu, Maruteru Masoori.

BPH-Prone Areas: Maruteru Samba & Masoori, MTU Rice -1271, Bapatla Masoori, BPT Rice- 2846, Parthiva, Nandyala Sona.

Late Planting: Samba Masoori, Nandyala Sannalu, MTU Rice-1293, NLR Rice -3238, Nellore Masoori, Cotton Dora Sannalu, Chandra.

Fertilizer Application: Grow green manure crops (Dhaincha, Sunhemp, Pillipesara and Cowpea) as preceding crop to Paddy and apply soil test-based fertilizers. The entire dose of phosphorus fertilizer should be applied during the final puddling stage. In heavy soils, the full dose of potash fertilizer should be applied at the final puddling. In light soils, half the dose should be applied during puddling and the remaining half at the panicle initiation stage.

Weed Control: In wet direct-seeded rice fields, within 3-5 days of sowing, apply Pyrazosulfuron-ethyl 10% WP @ 250 g/ha or use Pretilachlor + Safener 37% EW @ 1500 ml/ha mixed with 50 kg of dry sand and broadcast evenly over the field. In transplanted paddy fields, within 3-5 days of planting, when the field has shallow standing water, apply either Pyrazosulfuron-ethyl 10% WP @ 200 g/ha mixed with 50 kg sand, or Butachlor 50% EC @ 2.5-3.12 litres/ha, or Pretilachlor 50% EC 1000 ml/ha. If the field has a heavy infestation of grassy, sedge and broadleaf weeds, use Bensulfuron Methyl 0.6% + Pretilachlor 6.0% Granules @ 10 kg/ha mixed with 50 kg sand and broadcast evenly. At 20 days after transplanting, for grassy weed control, apply Cyhalofop-P-butyl 10% EC @ 1000 ml/ha. If weed infestation is high, apply Bispyribac Sodium 10% SL @ 250 ml/ha or Triafamone + Ethoxysulfuron 30% WG at 225 g/ha in 500 liters of water.

False smut: Spraying of propiconazole @ 200ml/acre two times at booting stage and at 50% panicle emergence stage. At severe incidence of diseases spray tebuconazole + trifloxystrobin 75WG@80g/acre.

Bacterial leaf blight: Affected stubbles are to be destroyed by burning or through ploughing, judicious use of nitrogenous fertilizers, avoid clipping of tip of seedling at the time of transplanting. Avoid flooded conditions or drying of the field (not at the time of flowering). Application of potassium two time @ 15 kg/acre at planting time and 20 kg/acre at panicle initiation stage.

Trap Barrier System (TBS): Erect 2 ft height polythene sheet barrier around the crop at 15 days after transplanting. The polythene barrier is provided with 3 inch radius holes above the water level at 15-30 m distance and wonder traps are arranged firmly inside the crop, their entry mouths matching the holes made in the polythene sheet. Arrange a soil moat before the entry hole to facilitate the animal to enter easily inside the trap. Maintain water around the TBS and collect the successful traps every day and replace with new / emptied traps.

Precautions during natural calamities: If the crop is submerged in water for more than 1 to 2 days in the boot leaf or flowering stage, the panicle exertion may be incomplete. Water accumulation in the spikelets, would also result in reduced fertilization and formation of chaffy seeds. Spraying with Carbendiazim @ 1g/L or Carbendiazim + Mancozeb @ 2g/L or Propiconazole @1ml/L is recommended to avoid discoloration of grains. If the harvested sheaves in the field are drenched, 5% salt solution should be sprayed on the sheaves to prevent seed germination. The sheaves should be turned over, dried and threshed after cessation of the rains. In

must. Avoid water stagnation at all stages. Apply KNO_3 or 19:19:19 @ 1.0 per cent as foliar spray to mitigate moisture stress. To correct iron deficiency, spray FeSO_4 @ 0.5 per cent twice at 7-10 days interval.

COMMERCIAL CROPS

Cotton

Sowing during June to mid-July in red soils and July to mid-August in black soils. Avoid light red soils without irrigation facilities. Sowing Non-Bt cotton/redgram around the Bt cotton as refugee crop to avoid build-up of resistance against bollworms. Use of pre-emergence herbicide Pendimethaline @ 1 L/acre immediately after sowing and post emergence herbicides Quizalofop-p ethyl 400ml+ Pyriproxyfen sodium 200 ml/acre at 25-30 DAS. Use mechanical weeders at early stages of crop. Adopt drip irrigation for higher yield and water saving. Spraying 10 g MgSO_4 /l to manage Mg deficiency at 45 and 70 DAS. Spray 2 g of ZnSO_4 /l 2-3 times at 5-6 days interval to manage Zn deficiency. Applying 20 kg ZnSO_4 per acre in soil once in three years. Spray 1-1.5 g borax/l at 60-90 DAS twice at weekly intervals to manage boron deficiency. Spray 2 % urea or 1 % 19:19:19 or 20:20:20 or 21:21:21 or 13-0-45 (KNO_3) to manage drought or stress conditions. Under submergence due to excess rains, drain the excess water at the earliest, apply 20+10 kg N and K/ha, loosen soil, spray 1 % KNO_3 or 19:19:19 or 20:20:20 or 21:21:21. Drenching of affected plants/rows with a mixture of Copper oxychloride 25 g + 200 g Urea in 10 liters of water immediately after occurrence of Parawilt within 48 h can revive the plants. Use yellow and blue sticky traps @ 10 /acre to control the sucking pests. Foliar application of Salicylic acid/ phytohormones - Salicylic acid improves photosynthetic capacity, promote root architecture and enhance plant defence mechanisms, stress tolerance, and overall growth. Using Drone: Spray Salicylic acid @ 6.9 g dissolved in 300 ml of ethyl alcohol and then added into 20 litres of water for spraying of one acre at 5 days after waterlogging. Three days after salicylic acid application, spray KNO_3 (13:0:45) @ 400 g of dissolved in 20 liters of water through drone. Knapsack sprayer: Spray Salicylic acid @ 6.9 g dissolved in 300 ml of ethyl alcohol and then added into 200 litres of water for spraying of one acre at 5 days after waterlogging followed by 2 % of KNO_3 (13:0:45) after three days of salicylic acid application.

Sugarcane

Varieties

Early maturing: 84A 125, 87A 298, 86V 96, 2003V 46, 93A 145, 97A 85, 2001 A63, 2003A 255, 2009V 127 (Ranga), 2005T 16, 2009A 107 and 2012A 319.

Mid-late maturing: CoT 8201, CoA 7602, Co 7805, 98 A 163, 2000 A 225 and 2009A 252. *Late maturing:* Co 7219.

Moisture stress: Co T 8201, 87 A 298, 97 A 85, Co 7219, Co A 7602, 98A 163 and 2005T 16.

Saline - Alkaline soils: CoT 8201, Co 7219, 97A 85, 93A 145, 99 V 30, 2005T 16, 2003V 46, 2009V 127, 2009A 107 and 2009A 252.

requirement and Cost of the implement is Rs.1200/-. Cotton Picker mechanically plucks open cotton bolls from the plant without damaging the fiber. Used in large-scale operations.

Fruit Crops

Secateurs are used during June to August for pruning mango branches lesser than 1 inch thickness and for removing dry twigs. Cost of the falcon company implement is Rs. 450. Loppers are used during June to August for pruning mango branches and dry twigs less than 3 inches thickness. Cost of the falcon company implement is Rs. 1500. Mango harvester is used to harvest mangoes without damaging the fruits. Cost of the implement is Rs. 3150. Banana bunch cutter is used to harvest banana bunches. Coconut climbing cycles are used to easily climb coconut trees for harvesting coconuts and fronds and for pest and disease management. Cost of the implement is Rs. 3500/-. Coconut dehusker is used to remove the husk from coconut shells. Cost of the implement is Rs. 950. Oil palm harvester is used to harvest oil palm bunches, bring them down, and load them into a tractor.

Livestock

A milking machine is beneficial for farmers with more than 4-5 dairy cows as it extracts milk 3-4 times faster than manual milking and reduces physical labour. Farmers rearing more than 30-50 backyard poultry birds can use a vaccinator to administer the specified dose of vaccine in a very short time. When raising backyard poultry birds in large numbers, de-beaking with the help of a debeaker is a good management practice to reduce pecking. With the help of a duplex poultry shed, 10 - 15 chickens can be kept in a protective shelter providing with integrated feed and protection, and manage very easily at very low cost.

Drone Technology

Spraying of pesticides through Drones reduces time and cost of application and it is the Measure against labour scarcity. In Paddy, Maize, Jowar, Groundnut, Bengalgram, Chilli crops, this technology is already proved effective. An Acre can be sprayed with 10 litres capacity of Drone in 6-10 minutes.

ENTERPRISES

Adopt secondary agriculture through value addition to primary agriculture for enhancing farm income and sustainable rural livelihoods. Agripreneurship promotion through farm mechanization, crop diversification, processing, value addition, organic farming. Take up Agro based enterprises such as commercial horticulture, cultivation of high value crops, animal husbandry, fisheries, poultry, and Mushroom cultivation for doubling farm income. Setting up of Agro produce processing units like Mini dhal mills, Rice Mills and decorticating mills for promotion of agripreneurship among rural youth & women. Value addition of agriculture producing through Agro produce manufacturing units such as preparation of millet value added products, bakery products, ready to cook products, ready to eat products and nutraceuticals. Entrepreneurship by agro- input manufacturing units such as production of organic manures, fertilizers, agricultural inputs, vermin compost units, Bio fertilizers and agricultural implements. Establishments and

Anthrachnose in mango and papaya use Carbendazim 50% WP@1 g/L spray during flowering and fruit set stage however it is also sprayed during rainy season if spots appear. To control Mealybug in papaya, use Chlorpyrifos 20% EC @2.5 ml/L which is sprayed on base and crown after rains. To control scale insect in guava, apply Buprofezin 25% SC @1 ml/L and spray on affected trees twice at 15-day interval. For stem borer, clean the hole then soak cotton bolls in chlorpyrifos @5 ml/L water or Dichlorvos @5 ml/L water and place in it and then seal the hole with mud. In citrus fruit, use Imidacloprid 17.8% SL @0.3 ml/L spray at new flush stage for Leaf Miner.

In Kharif season farmers are advised to grow vegetables like Okra, Brinjal, Tomato, Cucurbits. Ridge and furrow method of cultivation is recommended for vegetable cultivation along with proper drainage of water. Farmers are advised to sow okra seeds at a depth of 1-2 cm, with row-to-row spacing of 45 cm and plant-to-plant distance of 15-20 cm. Recommended varieties for okra cultivation are Kashi Kranti, Kashi Satdhari, and Kashi Chaman. For tomato cultivation, ensure planting on raised beds and provide staking using metal wires, nylon ropes, or bamboo supports, especially when cultivating indeterminate varieties such as Arka Rakshak, Arka Samrat, Swarna Anmol, and Swarna Samridhi. Determinate varieties like Swarna Sampada and Swarna Prakash should also be included for a balanced and productive harvest. Suitable varieties of cucurbits such as bottle gourd, bitter gourd, cucumber, sponge gourd, ridge gourd, snake gourd, and spine gourd should be sown with appropriate spacing and the recommended dose of fertilizers to ensure healthy growth.

To support vigorous vine development, a sturdy trellis structure made of bamboo and wire should be constructed which helps in allowing the plants to spread efficiently and enhance air circulation and fruit quality. To control Yellow Vein Mosaic Disease in okra, farmers should remove and destroy affected plants as soon as symptoms appear. After that, spray Thiamethoxam 25 WG at 100 g per hectare for effective disease management. Wilting in brinjal and tomato plants can be controlled by spraying of 1g Streptocycline+30 g copper oxychloride in 30 litre of water in the root zone of the plants. To control red pumpkin beetle in cucurbitaceous crops, apply chlorpyrifos 2% dust @20 kg/ha near the root zone and mix them properly.

COMMERCIAL CROPS

Sugarcane

From June onwards, sugarcane enters its early vegetative and grand growth phase, especially for spring-planted (February-March) crops, while late planting (autumn planting) can still be done until early July where irrigation or pre-monsoon rains are available. Ensure earthing-up of spring-planted crops by mid-June to promote root growth, prevent lodging and enhance moisture conservation. This is also the right time for trash mulching to suppress weeds, conserve soil moisture, and support beneficial microbial activity. Any gap filling should be completed early in June using healthy setts to maintain a uniform crop stand. Nutrient management during this period is crucial. For previously applied basal dose followed by the second split of nitrogen (around 60-75 kg N/ha) using urea before rains or irrigation. Intercropping

Nutrient management includes application of potassium at 50 kg/ha along with recommended nitrogen and phosphorus doses, and in saline soils, gypsum should be applied at 5 t/ha. To enhance crop nutrition, foliar spray of 0.2% boric acid or nano boron should be applied at 30, 45, and 60 DAS. For plant protection, collar and stem rot can be managed with a spray of Mancozeb or neem extract around 75 DAS. White grub infestation can be controlled through seed treatment with Chlorpyrifos or Imidacloprid, and soil application of *Metarhizium anisopliae* or Carbofuran. Sucking pests should be managed through seed treatment with Imidacloprid or foliar spray with Thiamethoxam, while defoliators can be controlled using Quinalphos, Chlorpyrifos, or Flubendiamide. Harvesting should be done at 120-125 DAS using a blade harrow, followed by threshing to separate the pods.



Groundnut Demo Plot

Soybean

Soybean grows best in well-drained loamy soils and should be avoided in saline or alkaline conditions. Deep ploughing and the application of farmyard manure (FYM) are essential for soil preparation. Recommended varieties include JS 335, GJS-3, and GS-4. Sowing is done in June-July with a spacing of 45×5 cm and a depth of 3-5 cm, using a seed rate of 70-80 kg/ha. Seeds should be treated with Carbendazim or Thiram along with *Rhizobium* and phosphate-solubilizing bacteria (PSB) to enhance germination and nutrient uptake.

For weed management, Pendimethalin is applied pre-emergence and Imazethapyr post-emergence, along with hand weeding at 20 and 40 days after sowing (DAS). To manage sucking pests, seed treatment with Imidacloprid is effective, while collar rot can be controlled by treating seeds with Fluxapyroxad or a combination of Carboxin and Thiram. In case of defoliator infestation, spraying Indoxacarb, Lambda-cyhalothrin, or a Novaluron mixture is recommended. For rust and leaf spot diseases, fungicides like Hexaconazole or Pyraclostrobin should be applied. Additionally, 75% of the recommended dose of fertilizers (RDF) should be supplemented with 25 kg/ha of Mycorrhiza and *Rhizobium* seed treatment to enhance soil health and crop productivity.

Cotton (Bt & Desi)

Cotton cultivation thrives best in well-drained loamy to sandy loam soils. Prior to sowing, deep ploughing is essential along with the incorporation of 10-15 t/ha of farmyard manure (FYM) to enhance soil fertility. Sowing is typically carried out in June-July under rainfed conditions and during April-May for irrigated crops. Recommended spacing for normal planting is 90-120 × 30-45 cm, while under high-density planting (HDP) systems, 60 × 30 cm spacing is advised. Suitable varieties include Bt hybrids like G. Cot. Hy. 24 and 26, and desi types such as GADC-3 and GADC-4. The seed rate varies by type, with 1.5-2 kg/ha for Bt varieties and 10-12 kg/ha for desi types.

Vaishali, while late maturing ones are BDN-2, Anand Grain Tur 2, and IPCL-87119. Sowing should begin as soon as sufficient rainfall is received in June-July. Land preparation involves applying 8-10 t/ha manure followed by light deep ploughing and 1-2 harrowings to ensure uniform and level soil. Spacing for early maturing varieties should be 45-60 × 15-20 cm, while for late maturing types, wider spacing of 90-120 × 25-30 cm is recommended. Fertilizer application should follow a dose of 25:50:00 kg/ha of N:P:K. Gap filling and thinning should be carried out 10-15 days after sowing (DAS).

For managing *Helicoverpa armigera*, planting yellow marigold around and between rows is effective. Sterility mosaic disease, which is mite-transmitted, can be controlled by spraying Propargite, Fenazaquin, or Fenpyroximate at 10 ml per 10 liters of water. To manage powdery mildew, spray soluble sulfur (80% WP) at 30 gm or Carbendazim, Hexaconazole, or Dinocap at 10 ml per 10 liters of water.

Coconut

For hybrid (D × T) palms, apply 270 g of urea/plant/month through drip irrigation. The drip schedule should be adjusted seasonally, with 1.5 hours of irrigation/day from October to February and 2.5 hours per day from March onward. Each palm should be equipped with four drippers, each delivering 8 liters/hour, positioned approximately 1 m away from the trunk to ensure even and efficient water and nutrient distribution.

SOUTH GUJARAT

This region of Gujarat includes Surat, Navsari, Valsad, Tapi, Bharuch and Narmada, SAU: Navsari Agricultural University, Navsari (Gujarat). These area experiences rainfall ranging from 400 to 900 mm annually. The soil is predominantly deep black cotton soil, which is known for its fertility and suitability for various agricultural activities.

Paddy

Select high-yielding or hybrid varieties suitable for South Gujarat such as GNR-3, GR-11, GR-7, IR-28, and hybrid PRH-10. Prefer puddled lowland fields with good water retention and proper bunding for transplanting. Raise healthy nursery on well-prepared land; sow 20-25 kg seed/ha for varieties and 10-12 kg/ha for hybrids. Treat seeds with *Trichoderma viride* or Bavistin to prevent seed-borne diseases. Apply FYM @ 5-10 t/ha and basal NPK @ 100:50:50 kg/ha depending on soil test recommendations.

Transplant 21-25 days old seedlings, using 2-3 seedlings per hill at a spacing of 20×15 cm. Maintain 2-5 cm standing water in field for 15-20 days post-transplanting, then adopt alternate wetting and drying. Apply pre-emergence herbicide Butachlor @ 1.5 L/ha 3 days after transplanting; do hand weeding at 30 and 45 DAT. Apply top dressing of nitrogen in 2-3 splits (at tillering, panicle initiation, and flowering stages). Monitor for stem borer, leaf folder, and gall midge; use chlorantraniliprole or fipronil as needed. For sheath blight and blast, apply tricyclazole or validamycin at recommended doses. Maintain field sanitation; drain excess water to reduce disease and pest incidence.

Stop irrigation 10-12 days before harvesting; harvest when 80-85% grains in the panicle are matured. Dry harvested grains properly to maintain quality and reduce post-harvest losses.

Groundnut

Use improved groundnut varieties like GJG-32 and GJG-22 with a seed rate of 100-120 kg/ha. Treat seeds with *Rhizobium* culture and appropriate insecticides to prevent white grub infestation. After the first rainfall, spray surrounding trees to control grub sources and minimize field infestation.

Sesame

For effective cultivation, land preparation should focus on creating well-drained, loamy soil through 1-2 ploughings followed by levelling. The ideal sowing period is from the last week of June to mid-July, immediately after the onset of the monsoon. Recommended varieties include Gujarat Til 1, Gujarat Til 2, and Gujarat Til 3. The seed rate should be 5-6 kg per hectare, with a recommended spacing of 30 cm × 10 cm (row × plant) to ensure optimal growth. Seeds should be treated with either *Trichoderma viride* at 4 g/kg seed or Carbendazim at 2 g/kg seed to prevent seed-borne diseases.

For fertilizer application, a basal dose of 40 kg N and 20 kg P₂O₅ per hectare should be applied at the time of sowing. Weed management can be achieved through interculture or hand weeding, which should be done twice-once at 15-20 days after sowing (DAS) and again at 30-35 DAS. Pest management for leaf roller or capsule borer involves spraying Neem oil at 2% or Spinosad 45 SC at 0.3 ml/L. To manage diseases like Alternaria leaf spot or Phyllody, infected plants should be removed and, if necessary, Mancozeb 0.2% should be applied. Water management is primarily rainfed, but light irrigation should be provided if there is a dry spell lasting more than 10 days, particularly during the flowering or pod-filling stages.

Cotton

For *Bt* cotton cultivation, select recommended hybrids suited to the region, such as GCHH 245, GCHH 177, RCH 2 BG-II, and JKCH 1947. The ideal sowing period is from mid-June to early July, coinciding with the onset of the monsoon. Land preparation should involve deep ploughing followed by harrowing to achieve good tilth. Before sowing, treat the seeds with Carbendazim or Thiram at 2 g/kg to prevent seed-borne diseases.



Pest Surveillance in Cotton

The recommended spacing for *Bt* cotton hybrids is 120 cm × 60 cm, ensuring a proper plant population of approximately 13,000-15,000 plants per hectare. Fertilizer application should include 240 kg of nitrogen, 60 kg of phosphorus, and 60 kg of potassium per hectare. Apply 25% of the nitrogen, along with the full dose of phosphorus and potassium, as the basal dose, and split the remaining nitrogen into 2-3 applications during the growing

Plant protections: For the management of pod borer application of chlorantraniliprole 18.5 SC @ 150 ml/ha followed by flubendiamide 480 SC @ 125 ml/ha is advised to minimize the larval population or Spinosad (150 mL/ha) at flowering and pod initiation. In incidence of blister beetles deep ploughing, field sanitation, light traps, and neem-based products (NSKE 5% or Azadirachtin 0.03% WSP @ 2500-5000 g/ha) can be used.

Disease management: Sterility mosaic virus can be controlled by using resistant varieties and removing infected plants early to limit virus spread. For vector control, spray Neem oil (5 ml/litre) or NSKE 5% every 15 days. Seed treatment with Imidacloprid 70 WS (5 g/kg seed) helps protect seedlings. Alternaria blight can be managed by spraying Mancozeb (2.5 g/litre) or Carbendazim (1 g/litre), planting on raised beds for better drainage, and using resistant cultivars. Phytophthora blight can be controlled by seed treatment with Metalaxyl (8g/kg), maintaining good drainage, minimizing stem injury, and using resistant genotypes.

Summer moong

If infestation of Jassid and white fly is observed, apply 1.0 litre malathion 50 EC or 625 dimethioate 30 EC per hectare in 625 litres of water. Harvest the crop at ripening of 70-80 percent pods to allow timely sowing of the kharif crop.

OILSEED CROPS

Groundnut

Improved Varieties: MH-4, HNG-69, GJG-19.

Time of sowing: Optimum time of sowing of groundnut is month of June. Under irrigated conditions, sowing should be completed by second fortnight of June, whereas under rainfed conditions sowing at the onset of monsoon should be done. Sowing of crop after 15 July should be avoided.

Seed rate: The seed rate depends on the size of kernels of the varieties. However, for medium size kernels varieties, seed rate of 80 kg per hectare is required.

Method of sowing: Sowing should be done preferably in rows at a spacing of 30 cm and maintaining plant to plant spacing of 15 cm.

Seed treatment: Treatment of seed with Thiram or Captan or Emisan 3 g/kg seed is essential for management of seed and soil borne diseases. For management of termites and white grub treat the seed with 15 ml Chloropyriphos 20 EC/kg seed.

Fertilizer use: Application of 15 kg N, 50 kg P₂O₅, 25 kg K₂O and 25 kg Zinc sulphate per hectare is required for obtaining good crop yield. It will be desirable to apply doses of all the nutrients before sowing. Use of gypsum as sulphur source has been found beneficial in groundnut crop.

COMMERCIAL CROP

Cotton

Time of sowing: Sowing of the crop is done from April to first fortnight of June. However, in case of American cotton optimum time of sowing is May. Delay in sowing

results in yield reduction. For all desi cotton varieties/hybrids, best sowing time is mid-April to 1st week of May. The mortality of seedling is very high during May & June.

Varieties and hybrids: Improved American cotton varieties: HS-6, H-1117, H-1126, H-1098 Improved, H-1236, H-1300. Recommended improved hybrids- HHH-223, HHH-287. Desi Cotton varieties- HD-123, HD-324, HD-432. Desi Cotton hybrid - AAH-1

Seed treatment and seed soaking: Before sowing, seed should be dipped in water up to 5-6 hours for better germination. Treat the seed with 5 gm Emisan, 1 gm streptomycin and 1 gm succinic acid in 10 litres of water. In termite affected areas treat the seed with 10 ml chlorpyrifos apart from above mentioned chemicals. Seed treatment with carbendazim @ 2 gm/kg in the root rot affected areas is essential. Seed treatment with Imidacloprid @ 7.5 gm/kg seed to escape the crops from sucking pests up to 40-60 days.

Seed rate: Use 15-20 kg delinted seed of improved American cotton varieties for sowing in one ha area. 12.5 kg of seed/ha is required for Desi cotton varieties. Seed rate of 3-3.750 kg /ha is required for American cotton hybrids and desi cotton hybrids. Seed rate of 2.125 kg/ha is required for Bt Cotton hybrids.

Method of Sowing: Sow the crop in lines 67.5 cm apart with a cotton sowing drill or cotton planter and plant to plant spacing of 60 cm or row to row spacing of 100cm and plant to plant spacing of 45 cm. Sowing should be done at a depth of 4-5 cm.

Fertilizer use: Apply 80-100 Kg N per ha for American Cotton and 50 kg N for desi cotton along with 30 kg P₂O₅/ha. In case of hybrid cotton, 150 kg N, 60 kg P₂O₅, 60 kg K₂O and 25 kg ZnSO₄ per ha has been recommended. Apply 1/3 dose of N, full dose of P₂O₅, full dose of potash and full dose of ZnSO₄ at time of sowing. Remaining dose of N should be applied in two equal splits at square formation and flowering stage. In sandy soil the research results have revealed that 90 per cent dose of fertilizer through soil application and 10 per cent through foliar spray at the boll development stage gave the highest seed cotton yield. Inoculation of cotton seed with C2, M4 and *Azospirillum* culture resulted into saving of 25-27 kg N/ha.

Plant protection: Leaf sucking pest & Leaf Curl Virus disease: seed should be treated with 4 g Thiamethaxom 70 WS/kg seed before sowing. Close monitoring on white fly host plants like vegetables, flower plants, weeds & unwanted plants. These host plants must be uprooted & burnt from time to time. Some vegetables like Ladies Finger, Brinjal, Tomato & Chilli also act as host plants for whitefly, accordingly, as per need these must be sprayed with Thiamethaxom 25 WG @ 0.5g/litre water. Bacterial blight: seed should be treated with soaking in 1 g Streptomycin or 10 g Plantomycin/10 litre water before sowing. Spray Thiamethaxom 25 WG @ 100g/ha and Profenophos 50 EC @1250 ml/ha for sucking pest. Proclaim (emamectin benzoate) 5 SG @250g/ha for pink, spotted and American bollworm.

FRUIT & VEGETABLE CROPS

Mango

In horticultural crops at fruiting stage such as mango, while carrying out field operations related to nutrient sprays and crop protection adequate precautions in handling of inputs,

mixing, delivery and washing of equipment is to be undertaken. For management of fruit drop in mango spray 2% urea and 0.5 % ZnSO₄ or 20 ppm 2, 4-D (2 g 2,4-D in 100 litre of water) in month of April-May. For management of black tip in mango apply 0.6% borax. For management of mango Jassid, spray 1250 ml malathion 50 EC in 1250 litre water per hectare. Manage irrigation in mango, citrus, and other fruit crops.

Citrus

In Citrus plants of seven years or more age, apply 750 g of urea per plant followed by inter-culture and irrigation. To control pre-harvest fruit drop in citrus, apply 6 g 2, 4-D, 3 Kg Zinc Sulphate, 12g Aureofungin and 1.5 kg lime in 550 litres of water per acre. In case of intercropping with sunflower and cotton do not spray 2, 4-D. In this case application of 20 ppm NAA is recommended.

Ber

In case of Ber plants last week April month to May month is best time for pruning because plants are in dormant condition.

Keeping in view of forecasted high temperature, farmers are advised for harvesting of mature vegetables at morning and evening hours and keep harvested crops in shadow. Optimum moisture level should be maintained in cucurbitaceous crops by light and frequent irrigation as dry conditions may lead to poor pollination and thus drop in yield of the crop. After harvesting of mature okra application of urea @ 5-10 kg/ acre should be done. Constant monitoring of crop against attack of mite is advised. If population is above ETL then, spraying of Ethion @ 1.5-2 ml/litre of water is advised. Due to prevailing high temperature, light irrigation at short interval is advised. To control shoot and fruit borer in brinjal and tomato crops, infested fruits and shoots should be collected and buried inside the soil. If pest population is high, spraying of Spinosad 48 EC @ 1 ml /4 litres of water is advised. In tomato, brinjal, chillies, bhendi, and other vegetables, apply hoeing and irrigation as needed and give the required amount of nitrogen fertilizer. Farmers are also advised to take short duration vegetable crops during this summer and the public to grow vegetable in their kitchen garden.

Cluster bean

Time of Sowing: Under rainfed conditions, sowing should coincide with the onset of the monsoon, ideally within the first two weeks of July.

Seed rate & spacing: Seed rate of 15 kg/ha is recommended with spacing of 30 cm x 15 cm, prior to sowing treat seeds with *Rhizobium* culture. Also treat seed with streptomycin @ 6 g/kg seed in 6 L of water.

Improved varieties: HG 365, HG 563, HG 870, HG 2-20

Fertilizer management: Apply 40 kg phosphorus and 20 kg Nitrogen at the time of sowing.

Water management: It is mainly grown as a rainfed crop; lifesaving irrigation at pod formation is critical.

Weed management: Two manual weeding at 25 and 45 DAS are usually sufficient. To control weed Basaline @2.0 litre per ha in 650 litre of water at pre-sowing can also be applied.

t/ha and incorporate during land preparation. Apply the following as basal dose during the final ploughing: Urea: 15 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha. To control early-stage weed infestation, apply pre-emergence spray of Pendimethalin @ 1 kg a.i./ ha within 2-3 days after sowing (DAS). Recommended Varieties: Shalimar Moong-1 & Shalimar Moong-2

Rajmash (Kidney Bean)

Complete sowing in the first fortnight of May. Bush type: Seed rate: 60-70 kg/ha & Spacing: 30 × 10 cm. Pole type: Seed rate: 30-35 kg/ha & Spacing: 60 × 20 cm. Recommended Varieties: Bush Type: Shalimar Rajmash-1, Shalimar Rajmash-2 and Pole Type: Shalimar Rajmash-3, Shalimar Rajmash-4. All other practices are the same as those for moong.

Cowpea

Complete sowing in the first fortnight of May. Use a seed rate of 40 kg/ha. Maintain a spacing of 30 × 10 cm. Recommended Varieties: Shalimar Cowpea-1 & Shalimar Cowpea-2. All other practices are the same as those for moong.

Soybean

Complete sowing in the first fortnight of May. Use 40-50 kg seeds/ha. Maintain a spacing of 45 × 10 cm. Recommended Varieties: Shalimar Soybean-1 & Shalimar Soybean-2. All other practices are the same as those for moong.

DISEASES OF PULSES

Diseases of Rajmash, moong, cowpea: Symptoms: The infected plants show gradual wilting and drying. Yellowing and premature drying of leaves and stems are noticed. The basal portion of stem and root region becomes black. When the bark of infected root is peeled, black streaks are seen and vascular tissues show discolouration.

Leafspot: Symptoms: Leaves show light brown small spots. Shot holes are formed in due course. Lesions develop on petioles and stem

Angular leafspot: Initial symptoms appear on leaves as circular spots in between the veins which attains angular shape which are dark grey in colour and latter turn to reddish brown to dark brown.

Anthracnose: Most striking symptoms appear on immature pods although symptoms also appear on leaves, petioles and stem. Infection appears as brick red colour on the veins on lower surface of the leaves. On veins brownish black lesions appear on pods sunken cankerious spots with lighter or grey centers develop.

Rust: The rust pustules appear on all the green parts. On the leaves there are numerous small, orange brown pustules each surrounded by a light-yellow halo. Infected leaves wither and may fall. similar rust pustules as on leaves develop on stem.

Bean Common mosaic virus: Downward curling of foliage is diagnostic character in bean common mosaic virus. The leaves of infected plants become mottled, chlorotic, puckered and tend to droop, leaf size is reduced.

White Mold: Appearance of light brown water-soaked discolouration on the stem, leaves or pods and cottony growth on in collar region under humid conditions.



In-situ moisture conservation measures covering both soil and land management practices to be practiced. They include field bunding, compartmental bunding, ridges and furrows, conservation furrows, broad bed and furrow systems, dead furrows at 30 DAS especially in red soils for groundnut, groundnut+redgram, finger millet + pigeonpea, pigeonpea + field bean and pigeonpea + cowpea intercropping. Frequent interculture required both in kharif and rabi for weed control and moisture conservation. Avoid dry sowing of crops such as cotton, groundnut especially in shallow soils. Sowing needs to be done only when sufficient moisture is available in soil root zone. Unless 50 - 60 mm rainfall is received over 2-3 days, sowing should not be carried out. Facility of crop insurance including weather-based insurance to be availed. Follow the weather advisories available for the region. Do not burn crop residue. Take up rotavator operation in harvested fields of Cotton, Wheat and Safflower for incorporation of stubbles and crop residue in the soil. Border crops (Jowar/Bajra/Maize) should be practiced in vegetable growing fields which will prevent the vector entry in the main field. Attend ploughing operation immediately after the harvest of rabi crops which reduces the pest & weed incidence in next season.

CEREAL CROPS

Paddy

Paddy is commonly cultivated using a seed rate of 62.5 to 75 kg per hectare, depending on the variety. Popular varieties include MO-4, Abhilasha, Intan, RNR 15048, Tunga, KHP-10, and BPT 5204. Sowing is generally done in June to August, with some varieties like RNR 15048 and BPT 5204 sown in the first fortnight of July and transplanted in August. The recommended fertilizer dose for most varieties is 75:75:90 kg of N:P:K per hectare, while BPT 5204 and RNR 15048 require a higher nitrogen input of 150 kg along with 75:75 kg of phosphorus and potassium. Seed treatment is a crucial practice and involves the use of Carbendazim 50 WP at 2 grams per kg of seed to prevent fungal infections. In some cases, seeds are also treated with

Control leaf spot with Mancozeb at 2 g/l + COC at 3 g/l, downy mildew with Sulphur at 3 g/l + Tebuconazole at 1 ml/l, and powdery mildew with Propiconazole or Hexaconazole at 1 ml/l. Manage yellow vein mosaic virus using *Trichoderma* at 4 g/kg seed and rogue out infected plants. For insect pests like sucking pests, spray Dimethoate at 1.7 ml/l + Acetamiprid at 0.3 g/l, Thiomethoxam at 0.3 g/l for thrips, and Chlorantraniliprole at 0.2 ml/l for pod borers. During periods of water stress, apply foliar spray of micronutrients at 2 g/lL

Black gram

Blackgram varieties such as DU-1 and TAU-1 are commonly sown in the month of June, using a seed rate of 15 kg per hectare. A basal dose of 25:50:0 kg of NPK per hectare is recommended, with the entire fertilizer quantity to be applied at the time of sowing. Phosphorus is preferably applied in the form of rock phosphate for better effectiveness. Prior to sowing, seeds should be dipped in a 2% calcium chloride (CaCl_2) solution for 30 minutes to help mitigate drought stress, followed by shade drying for 8 hours. Seed treatment involves the application of 500 grams each of *Rhizobium* and *phosphorus solubilizing bacteria (PSB)* per seed lot.

Blackgram should be sown in rows with a spacing of 30*10 cm. During the flowering stage, foliar application of Pulse Magic at 10 g/ litre of water is recommended to promote flowering and pod development. For weed control, Fluchloralin 45 EC should be applied at 2 L/ h2 as a pre-emergence herbicide. The crop is susceptible to powdery mildew, which can be managed by spraying Hexaconazole at 1 ml per litre of water. Thrips infestation can be controlled by spraying either Thiomethoxam at 0.2 grams or Imidacloprid at 0.3 m/L.

COMMERCIAL CROPS

Cotton

Cotton is the major crop of our district and commonly cultivated varieties include Prvardhan Reventha, Rasi Seeds RCH-929, and UAS Agriculture varieties. The crop is sown by dibbling method during the first fortnight of June with a spacing of 90 cm x 30 cm. The recommended fertilizer dose is 60:30:30 kg NPK/ha, and seeds are treated with zinc sulphate, iron sulphate, and manganese sulphate each @ 4 g/kg of seeds before sowing. Redgram is generally used as an intercrop. For foliar nutrition, magnesium sulphate @ 5 g/litre is sprayed. Being a rainfed crop, irrigation is generally not required, and weeding is carried out manually. For disease management, angular leaf spot is controlled by spraying carbendazim + mancozeb @ 2 g/litre, while thrips are managed by spraying imidacloprid @ 0.3 ml/litre, and pink bollworm (PBW) is controlled by profenofos @ 2 ml/litre.

Sugarcane

Sugarcane varieties suited for Belagavi include Co 86032, CoC 671, Co 94012, SNK 632, SNK-635, and SNK-08101, with the ideal sowing seasons being June-July and October-November. The seed requirement is 7,000 to 8,750 setts per hectare, and the recommended fertilizer application is 250:75:187.5 kg/ha of N:P:K for optimal growth. Setts should be treated before planting by soaking them in a solution of 100 g Carbendazim 50 WP, 100 ml Chlorpyrifos 20 EC, and 100 g urea in 100 liters of water for 10 minutes. The planting methods include ridges and furrows (spacing of

Chilli

Chilli cultivation includes both irrigated and rainfed practices, with popular varieties being Byadagi Dabbi, Byadagi Kaddi, Hybrid 5531, Krishnaprabha Rudra, Noolvi Dabbi for dry chilli, and Arka Tanvi, Arka Yashasvi, and Nagavi Giddagayi for green chilli. Under irrigated conditions, the seed rate is about 250 to 400 grams per hectare, whereas for rainfed conditions, it ranges from 1 to 1.25 kilograms per hectare. The ideal sowing time is May-June for kharif and October or January-February for irrigated crops, with transplanting carried out from July to August. Fertilizer recommendations for irrigated chilli are 150:75:75 kg NPK/ha, while for rainfed conditions, it is 100:50:50 kg NPK/ha.

Seed treatment should be done using Imidacloprid 70 WS at 10 g/kg, Thiram 75 WP at 2 g/kg, *Trichoderma viride* at 4 g/kg, or *Azospirillum* at 200 g/kg, and seedlings can be dipped in *Azospirillum* solution (400 g per litre of water) during transplanting. Spacing should be 75 cm x 45 cm under irrigation and 90 cm x 90 cm under rainfed conditions. Both transplanting and direct sowing methods are practiced. Intercropping with marigold and crops like onion or Bt cotton is encouraged, along with border cropping using hybrid maize or sorghum. A foliar spray of NAA at 0.3 ml/L during flowering and Arka Vegetable Special at 3 g/litre enhances plant growth and yield. Irrigation should be scheduled once in 5-7 days in red soil and once in 15-17 days in black soil.

Weed management is achieved using Pendimethalin at 0.5 litres/ha or 4 ml/litre applied on the day of sowing or the next day. Major diseases like anthracnose, powdery mildew, *Alternaria* leaf spot, and murda complex can be managed by integrated practices. Specific foliar sprays include Carbendazim 50 WP at 1 g/litre or Fenconazole 10 EC at 1 ml/litre for powdery mildew and Mancozeb 75 WP at 2 g/litre or Copper Oxychloride at 3 g/litre for anthracnose. Pests such as thrips, mites, and whiteflies are controlled through IPM practices including seed treatment with Imidacloprid, foliar sprays of Difenthiuron 50 WP at 1 g/litre, and installation of 25 blue and yellow sticky traps per hectare.

Tomato

Recommended varieties for tomato include Arka Rakshak, Arka Samrat, Arka Abhijith, Arka Vardhan, and Arka Vikas, with fruits weighing around 75-100 grams. Suitable sowing periods are June-July, October-November, and January-February. For nutrient management, hybrids require 250:250:250 kg/ha of NPK, while varieties require 115:100:60 kg/ha. Seeds are treated with *Trichoderma subtilis* @ 50 g/kg or Thiram @ 1.25 g/kg for protection against seed-borne diseases. The crop is planted on ridges and furrows (90 x 45 cm) or raised beds (75 x 45 cm) for rainy season crops. Foliar nutrition with Vegetable Special @ 3 g/l is applied fortnightly.

Irrigation is scheduled once in 5-7 days based on moisture and weather. Weed control is achieved using pre-emergent herbicides like Alachlor or Butachlor @ 1.5 litre/ha. Major diseases include damping off (managed by seed treatment or drenching with Metalaxyl or Mancozeb), Tomato Leaf Curl Virus and Spotted Wilt Virus (managed by vector control and removal of infected plants), early and late blight (controlled by sprays of Azoxystrobin, Copper Oxychloride, or Metiram), bacterial wilt (managed by soil drenching with Streptocycline and crop rotation),

as banana, cocoa, nutmeg, black pepper, or ginger. Fertilizer recommendation is 100:40:140 g NPK per plant annually, applied in two equal splits - first in September-October along with 12 kg organic manure and second in February (for irrigated) or June (for rainfed). Additionally, zinc sulfate and borax are applied at 5 g per plant during October-December. Irrigation is usually done through the basin method at 175-200 litres per palm per irrigation, adjusted as per season: every 7-8 days in November-December, every 6 days in January-February, and every 4-5 days from March to May. Drip irrigation is also practiced. Common diseases include bud rot, leaf spot, collar rot, crown rot, foot rot, and yellow leaf disease. Bud rot can be managed with sprays of 1% Bordeaux mixture, 0.3% Copper Oxychloride (COC), or 0.2% Metalaxyl + Mancozeb 72WP, applied before the onset of monsoon and repeated after 30-45 days. Pests such as spindle bug, Pentatomid bug, red palm weevil, mites, root grubs, inflorescence caterpillar, and ambrosia beetle affect the crop. Red mite infestations are managed by spraying wettable sulphur (80%) at 0.4%.

LIVESTOCK

To ensure the health and productivity of livestock during the kharif season, it is essential to implement appropriate management practices. Cattle, buffalo, sheep, and goats should be vaccinated against Hemorrhagic Septicemia, while sheep and goats also require vaccination against Enterotoxaemia. Deworming of all animals using Piperazine, Albendazole, or Fenbendazole is recommended to control internal parasites during the rainy season. Along with green fodder, daily feeding of dry fodder to cattle, buffalo, and sheep is necessary for better digestion and nutrient absorption. Grazing of maize and jowar seedlings should be strictly avoided, as these can cause cyanide poisoning, potentially leading to animal death.

Pesticides, insecticides, and fertilizers must be stored safely away from animals to prevent accidental toxicity. Farmers can prepare animal feed using locally available ingredients, such as cereals (maize, jowar, wheat) 35-40 kg, bran or polish of grains 30-35 kg, oil cakes (groundnut, safflower, cottonseed) 25-30 kg, mineral mixture 2 kg, and salt 0.5-1 kg to make 100 kg of cattle feed. To prevent the spread of Lumpy Skin Disease (LSD), which is a vector-borne illness, livestock should be protected from mosquitoes, house flies, and ticks by using mosquito nets, repellents, and washing animals with ectoparasitocidal drugs. Additionally, burning dry neem leaves in the morning and evening can help reduce mosquito bites. Farmers are also encouraged to cultivate improved fodder and crops during the Kharif season, utilizing the monsoon rains. Furthermore, all backyard poultry birds should be compulsorily vaccinated against Ranikhet Disease.

Fisheries

Maintain 6 feet water in the fish pond for proper growth and development of fishes. Remove unwanted plants and animals prior to release of fishes in the pond. As a supplementary food, floating pallets or groundnut cake (25%), paddy husk (40 %), sweet potato powder (10%), fish meal (24 %), vitamin and mineral mixture (1%) should be mixed and feed scientifically. Continue to feed fishes 3-5 % of body weight on daily basis. The suitable breeds for fresh water aquaculture are catla, rohu, mrigal, amur common carps, jayanti rohu, punggiasius, GIFT tilapia, grass carp, silver carp, murrel. Prior to release of fingerlings (15 days) mix cow dung (100 kg/100 m²) in the pond for plankton production.



The Kharif season, stretching from June to October, is a crucial agricultural period for the state of Maharashtra, significantly influencing the livelihoods of millions of farmers. Characterized by the onset of the southwest monsoon, this season supports the cultivation of key crops such as rice, cotton, soybeans, pulses, and millets. However, climate variability, erratic rainfall, and changing weather patterns pose continuous challenges to sustainable agricultural production. In the light of these challenges, the Kharif 2025 Agro-Advisory aims to provide timely, region-specific, and science-based recommendations to empower farmers with the knowledge needed to make informed decisions.

REGIONAL ADVISORY

Maharashtra is divided into several agroclimatic zones, each with unique characteristics influencing crop selection, sowing windows, and input management. The following region-wise advisories aim to support farmers with tailored recommendations for improved productivity and resilience.

VIDARBHA REGION

The Vidarbha region of Maharashtra is composed of Bhandara, Chandrapur, Gondia, Gadchiroli, Nagpur, Amravati, Akola, Yavatmal, Buldhana, Wardha, Washim and possesses state agricultural university viz., Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

Soil Testing & Preparation (All Crops)

Soil testing is crucial for effective fertilizer management, optimizing input use, and maintaining soil fertility. For proper soil sampling, the recommended depths are 15-20 cm for crops like sorghum, rice, groundnut, and wheat; 30 cm for cotton, sugarcane, and banana; and 30 cm for fruit trees, sampled from a distance of 1-1.5 feet from the trunk. Additionally, applying decomposers to crop residues like straw and leaves enhances organic matter content, improves soil structure, and boosts water retention, contributing to sustainable agricultural practices.

COMMERCIAL CROPS

Cotton

The cotton crop is sown during May-June. Land preparation includes deep ploughing every three years for rainfed areas and annually for irrigated ones, followed by two harrowings and one pre-sowing harrowing. FYM or compost is applied at 12-15 cartloads for rainfed and 20-25 cartloads or 5 t/ha for irrigated fields, with 2.5 t/ha vermicompost if available. Seeds are treated with Imidacloprid or Thiomethoxam (7.5 g/kg), Thiram or Captan (3 g/kg), and biofertilizers *Azotobacter* (25 g/kg) and PSB (20 g/kg). Sowing times vary: 20-30 May for irrigated, early June for dry sowing, and 15-30 June post-monsoon. For Bt/Hybrid (irrigated), spacing is 120x90 cm with a seed rate of 2-2.5 kg/ha, while Deshi improved (rainfed) is sown at 60x15 or 60x30 cm with 9-15 kg/ha. Intercropping with green gram, black gram, or soybean (1:1) or with sorghum and pigeon pea (6:1:2:1) is recommended. Gap filling and thinning are done within 3-5 days after emergence. Irrigated Bt requires 40:60:60 NPK, while Hybrids need 35:50:50 NPK, both applied in two splits. Rainfed varieties receive 20-30:20-30:20-30 NPK as a basal dose using the ring method. Delayed application reduces yield by 10-40%. Foliar sprays of 2% urea (flowering) and 2% DAP (boll stage) are advised. Drip fertigation involves staged application: 10% at sowing, 20% at 20 DAS, 25% each at 40 and 60 DAS, and 20% at 80 DAS, with phosphorus applied as basal. Weed control is crucial for the first 60 days, requiring 3-4 hoeings and 2-3 weedings. Moisture conservation is done by opening furrows at 40-60 DAS. Rainfed cotton benefits from 2-3 protective irrigations during flowering and bolling.



High Density planting in cotton

Common pests like Jassid, Thrips, Aphids, and Mites are managed with Dimethoate 30 EC (10 ml/10 L) or Acetamiprid 20 SP (1.5 g/10 L), while whiteflies are controlled using yellow sticky traps (5 per acre) and Dimethoate 30 EC (33 ml/10 L). Bollworms are managed with Quinalphos, Spinosad, or Cypermethrin, and pink bollworm with pheromone traps, Neemark 300 ppm, deep ploughing, and heat treatment. Thirty yellow sticky traps and 2-3 light or pheromone traps per gin are suggested. Biopesticides like Neem 5%, *Verticillium lecanii*, and *Beauveria bassiana* are recommended in humid weather.

Plant Protection: Bacterial leaf blight is managed by treating seeds with Carboxin (1 g) and Thiram (3 g/kg), grey mildew with wettable sulphur (25 g/10 L) or Carbendazim (10 g/10 L), Myrothecium with Copper oxychloride (0.3%), Carbendazim (0.1%), or Mancozeb (0.25%), and *Alternaria* with seed treatment and Mancozeb (0.2%) sprays every 20 days. Para wilt is managed with a drench of 25 g Copper oxychloride and 150 g urea in 10 L of water.

Plant Protection: Bacterial leaf blight is managed by treating seeds with Carboxin (1 g) and Thiram (3 g/kg), grey mildew with wettable sulphur (25 g/10 L) or Carbendazim (10 g/10 L), Myrothecium with Copper oxychloride (0.3%), Carbendazim (0.1%), or Mancozeb (0.25%), and *Alternaria* with seed treatment and Mancozeb (0.2%) sprays every 20 days. Para wilt is managed with a drench of 25 g Copper oxychloride and 150 g urea in 10 L of water.

Special recommendations include intercropping AKH-081 with green or black gram, spacing PKV DH-1 at 60x45 cm with 50:25:25 NPK (rainfed), and applying 2% urea (flowering) and 2% DAP (bolling) foliar sprays. For AKH-081 in shallow soils, 5 t/ha FYM and 25:12.5 N:P are advised. For pre-monsoon yield boost, 2.5 t FYM and 75:37.5:37.5 NPK/ha are applied.

**Maize observation in field**

sowing while remaining 50 % N should be applied after 30 DAS. For the control of Fall Army Worm, treat the seed with 3 g/ Thiram per kg of seed, spray 5 % NSKE withing 30 days of sowing or at 5 % infestation. In addition, spray Emamactin Benzoate 5 % @ 4 g/ 10 L of water at 10-20 % infested plants. Earthing up should be done during dry spell for in situ moisture conservation.

COMMERCIAL CROPS

Cotton

Cotton should be sown after onset of monsoons or after 1st June only for effective management of pink bollworm. Well decomposed FYM should be applied to cotton @ 10 ton/ha before sowing. Recommended fertilizer dose of 120:60:60 kg NPK/ha should be applied to rainfed cotton out of which 40% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 30 % N should be applied after 30 DAS and 30 % after 60 DAS. Use of high yielding, short duration cotton variety like CICR-14, CICR-19, PKV-081, Suraj BT, Rajat BT, NH-1901,1902,1903,1904 etc. For irrigated cotton apply 150:75:75 kg NPK/ha out of which 20% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 40 % N should be after 40 DAS and 30 % after 60 DAS. Closer spacing and High-Density Cotton Planting (HDPS) should be adopted on medium and light soils respectively for higher yields by keeping plating distance of 90 x 30 cm for medium soils and 90 x 15 cm for light soils which will accommodate approximately 14000 and 29000 plants per acre respectively. Canopy management should be done by cutting monopodia at 45-50 DAS and nipping the tip at 90 DAS supported with one or two sprays of growth retardants if required under expert guidance. Earthing up should be done in cotton during dry spell in Kharif season to reduce evaporation losses of soil moisture. Follow IPM package for the control of pests and diseases in cotton as given below. Select short duration and sucking pest tolerant variety for sowing

**Whitefly on cotton**

- a. Follow crop rotation avoid cotton sowing followed by tomato, chickpea, etc.
- b. Sowing of intercrop or mixed crop like maize, cowpea, Black gram, green gram and use of trap crop like marigold, castor, etc.
- c. Sowing of okra as a trap crop in cotton for the control of Pink Bollworm.
- d. Avoid plant growth promoters and pesticides during first three months and use botanical pesticides or biopesticides
- e. Spray of 5 % NSKE at 60 DAS
- f. Install 40 pheromone traps per ha for mass trapping of pink bollworm.

sowing. Seed of green gram and black gram should be treated with Thiram @ 2 g/ kg followed by seed inoculation of Rhizobium and PSB biofertilizers @ 10 ml / kg of seed. Green gram and black gram can be taken as intercrop in cotton at 1:1 ratio for higher yield and monetary returns.

FODDER CROPS

Sorghum

Improved varieties of Sorghum like CSV-20, CSV-23, CSV-27, PVK-801, PVK-809, CSH-25 (Parbhani Sainath), biofortified variety like Parbhani Shakti should be used for sowing. Seed should be treated with Thiram @ 3 g/ kg followed by seed inoculation of *Azotobacter* and PSB biofertilizers @ 10 ml / kg of seed. Kharif Sorghum should be sown at 45 x 12.5 cm spacing by using 7.5 kg seed rate per hectare. Recommended fertilizer dose of 80:40:40 kg NPK should be applied to Kharif Sorghum out of which 50 % N and full P and K should be given at the time of sowing while remaining 50 % N should be applied after 30 DAS. For the control of Stem Borer and Fall Army Worm, spray 5 % NSKE or Quinolphos 35 % EC @ 15 ml / 10 L of water within one week from sowing. Take another spray at the interval of 10 days if required.

Pearl Millet

Improved varieties like Shraddha, Saburi, AIMP-92901 and biofortified varieties like AHB-1200 Fe, AHB-1269 Fe should be used for higher yield. Recommended fertilizer dose of 60:30:30 kg NPK should be applied to Pearl Millet out of which 50 % N and full P and K should be given at the time of sowing while remaining 50 % N should be applied after 30 DAS. Seed should be treated with Thiram @ 3 g / kg of seed followed by Metalaxil 35 SD @ 5 g / kg of seed and then inoculate with *Azotobacter* and PSB biofertilizers @ 25 g / kg of seed.

COMMERCIAL CROPS

Sugarcane

Deep ploughing followed by harrowing and levelling; ensure proper drainage as the region is prone to waterlogging. Use high-yielding, disease-resistant varieties like Co-86032, Co-94012, Co-8014, Co-M-0265 (Phule-265), or Co-C-671. Ideal time for adsali (pre-kharif) planting is July-August; Kharif planting should be done in June-July. Use 3-budded setts from healthy, disease-free canes; dip in fungicide solution (e.g., Bavistin @1g/liter) before planting. Adopt 90 cm row-to-row and 30 cm plant-to-plant spacing for proper growth and aeration. Apply 250:115:115 kg NPK/ha in three splits (at planting, 45-60 DAS, and 90 DAS). Use organic manure or FYM @ 25-30 t/ha for better soil health. First irrigation immediately after planting, then every 8-10 days depending on rainfall and soil moisture. Perform 2-3 hand weeding/inter culturing at 30, 60, and 90 DAS. Use pre-emergence herbicides like Atrazine @ 1.5 kg/ha.

Pest & Disease Management:

Early shoot borer: Remove and destroy infested shoots; apply Chlorantraniliprole or Carbofuran granules in affected furrows.

Top shoot borer/ White grub: Apply *Trichoderma* or *Beauveria bassiana* in soil and spray Chlorpyrifos if needed.

Pokkah boing Disease spray the crop with Mancozeb 75 WP @ 30 g or Carbendazim 50 WP @ 10 g in 10 litre of water. For the management of Rust and Leaf spot diseases spray the crop with Mancozeb 75 WP @ 30 g or Tebuconazole 25 EC @ 10 ml in 10 litre water.

PULSE CROPS

Pigeon pea

Select proper sowing windows based on maturity groups. Early varieties can be sown in First fortnight of June and Medium/Late varieties in Second fortnight of June. Ensure flowering avoids low sunlight/cloudy weather to prevent poor pod formation. Use well-drained black cotton soil (pH 7.0-8.5). Adopt deep ploughing or BBF/Ridge & Furrow systems to improve drainage and aeration. Raised bed planting and appropriate row spacing (e.g., 4-5 ft) for better management and intercropping potential. Use quality seeds with appropriate seeding rates: Early: 20-25 kg/ha. Medium/Late: 15-20 kg/ha. Treat seeds with *Trichoderma viride*, Thiram + Carbendazim, and Rhizobium + PSB for disease prevention and better nitrogen fixation. Fertilization: Apply 25-30 kg N, 40-50 kg P₂O₅, 30 kg K₂O/ha at sowing. Incorporate Sulphur, Zinc, and Iron if soils are deficient. Follow soil test-based recommendations for precision nutrition. Irrigation & Drainage: Though drought-tolerant, apply 3 key irrigations during branching, flowering, and podding stages. Use ridge planting in poorly drained soils to avoid waterlogging. For weed Management: Pendimethalin (0.75-1 kg a.i./ha) as pre-emergence to suppress early weeds. Conduct two mechanical weedings before flowering (20-25 DAS & 45-50 DAS). Cropping and Intercropping Systems: Utilize inter-row space with short-duration crops: mung, urd, cowpea, , and recommended intercropping arwe Pigeon pea-Urd-Wheat, Pigeon pea-Sugarcane, Pigeon pea (early)-Potato-Urd bean.

Pest & Disease Management: Use resistant varieties (e.g., Asha, BSMR-853, JKM-189). Follow integrated pest management (IPM): pheromone traps, Neem products, timely chemical sprays (Emamectin, Indoxacarb).

Soybean

Soybean should be sown by the row-to-row spacing is 45 cm, and plant-to-plant spacing is 7 to 10 cm. A seed rate of 65 to 75 kg per hectare is sufficient, depending on the seed size and germination percentage.

Soybean should be treated with Thiram @ 3g/kg or *Tricoderma* @ 4 g/kg of seed inoculum like *Rhizobium* and PSB @ 100ml/10kg of seed before sowing. For higher production use new released high yielding varieties like Phule Sangam (KDS 726), Phule Kimaya (KDS 753), Phule Durva (KDS 992. Apply NPK @ 50:75:40 Kg/ha at the time of sowing. Use Broad bed furrow planter (BBF) for sowing. It will be beneficial during less rainfall situation and excessive rainfall. For chemical weed control, Pendimethalin at 1.0 kg active ingredient per hectare can be applied as a pre-emergence herbicide immediately after sowing. Imazethapyr at 75 grams active ingredient per hectare can be applied as a post-emergence herbicide at 15-20 days after sowing to control broadleaf and grassy weeds.

pheromone traps @ 18 - 20 per ha to control the fruit fly damage. For management of dieback spray zinc sulphate 2.0 kg + lime 1.0 kg in 100 L of water and keep fruit fly trap to control fruitfly or spray Spinosad 0.4 ml per liter of water or lambda cyhalothrin 10WP 1.25 g per L of water. For management of Mealybug spray profenophos 1 ml + acephate 1 g + 10 g or soap powder Nematodes bio-control using nematode egg parasitic fungus *Paecilomyces lilacinus* @25 grams along with farm yard manure @ 100 kg, neem cake @ 250 g. For management of Fruitfly: Collect and destroy fallen and infested fruits summer ploughing to expose pupa. Use methyl eugenol lure trap (25/ha) to monitor and kill adults of fruit flies. Prepare methyl eugenol and Malathion 50 EC mixture at 1:1 ratio (take 10 ml mixture/ trap). Insecticides: Malathion 50 EC 0.05%. Bait spray combining molasses or jaggery 10g/l and one of the insecticides, Malathion 50 EC 2 ml/l, dimethoate 30 EC 1ml/L, two rounds at fortnight interval before ripening of fruits.

Grapes

Maintain a well-aerated canopy to reduce humidity within the vineyard and improve air circulation. Monitor for any nutrient deficiencies, particularly potassium, and apply appropriate fertilizers as needed. Keep the orchard clean by removing weeds and debris to prevent fungal and bacterial diseases. Ensure proper drainage and prevent waterlogging. The vigorous varieties of grapes the tendrils and the twins may be pruned. Management of April pruning in grape orchard should be done as per availability of water. During rest period, provide only need based irrigation to protect the existing leaves from drying, and also contribute towards increasing the reserves of the vines through photosynthetic activity. Care should be taken to reduce/stop the water in case new growth is observed on the shoot.

KHANDESH/NORTHERN MAHARASHTRA

This region of Maharashtra is comprised of Jalgaon, Dhule, Nandurbar, Nashik). Rainfall of the region is 600-900 mm and it has Medium to deep black soils. This region has state agricultural university viz., Mahatma Phule Krishi Vidyapeeth, Rahuri, Ahmednagar (M.S)

COMMERCIAL CROPS

Cotton

Use medium to deep black cotton soils with good drainage for optimal crop performance. Prepare land with 1-2 deep ploughings followed by harrowing; adopt ridge-furrow or broad bed-furrow method. Treat seeds with *Trichoderma viride* or Carbendazim for fungal diseases and imidacloprid for sucking pest protection. Sow *Bt* cotton hybrids like RCH-2 BG II or NCS-855 BG II from mid-June to early July, depending on monsoon onset. Maintain spacing of 120 × 60 cm for hybrids and 90 × 45 cm for varieties; use seed rate of 450-500 g/acre. Apply basal dose of NPK @ 50:25:25 kg/ha based on soil test; supplement with FYM @ 5-10 t/ha. Apply zinc sulphate @ 25 kg/ha in zinc-deficient soils. Ensure field has proper drainage to prevent waterlogging, especially during early growth stages. Provide protective irrigation during long dry spells, particularly during flowering and boll development. Apply Pendimethalin @ 1.0 kg a.i./ha as pre-emergence herbicide;

follow with hand weeding at 20-25 DAS. Monitor for sucking pests (jassids, aphids, whiteflies); control using imidacloprid 0.3 ml/l or thiamethoxam 0.25 g/l.

Further, install 5 pheromone traps per hectare to monitor bollworms. If needed spray emamectin benzoate or flubendiamide as per threshold levels. Manage leaf spots and blights with mancozeb or copper oxychloride sprays as needed. Apply top dressing of nitrogen at 30 and 60 DAS for vigorous growth. Use plant growth regulators like NAA to minimize flower and boll drop, if observed. Harvest mature, dry bolls; avoid harvesting during or immediately after rain to prevent boll rotting.

PULSE CROPS

Soybean

The crop must be sown before 1st week of July, after good rain, with a spacing of 45×57cm; depth <4?cm. the seed must be treated with Carboxin + Thiram 30g/10kg + Thiamethoxam 100ml/10kg. Give a hoeing at 15-20 DAS. Apply water at branching (30-35 DAS) & flowering (40-45 DAS).

Pigeon pea

Recommended varieties for sowing include Phule Trupti, BDN-711, and Renuka, with planting done before the first week of July. Suggested spacing is 60×20 cm, 90×20 cm, or 180×30 cm, depending on field conditions. Treat seeds with Trichoderma (5 g/kg) or a combination of Carboxin and Thiram (4 g/kg), along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 25:50 kg N:P per hectare. The seed rate varies-12-15 kg/ha for normal spacing and 5-6 kg/ha for wider spacing. Intercropping is recommended with bajra, soybean, sorghum, cotton, moong, or urad for better land utilization. Irrigate at 30-35 days after sowing (DAS), 60-70 DAS, and during the pod-filling stage. Perform one hoeing at 15-20 DAS followed by hand weeding for effective weed control.

Black gram

The recommended seed rate is 15-20 kg/ha with a spacing of 30×10 cm. Seeds should be treated with Trichoderma at 5 g/kg along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 20:40 kg N:P per hectare. Intercropping with pigeon pea and other kharif crops is advised for better resource utilization. Irrigation should be provided at critical stages such as flowering and pod filling to ensure healthy growth and yield.

FRUIT & VEGETABLE CROPS

Pomegranate (Mrig Bahar)

Pomegranate cultivation during Mrig Bahar requires pruning to remove diseased branches, followed by an application of 1% Bordeaux mix. Pre-flowering fertilization includes 700 g urea, 1600 g SSP, and 400 g MOP per tree. Dusting with 4% copper dust or bleaching powder helps in disease prevention. Regular pest control with Azadirachtin or alternate insecticides every 15 days is crucial. At the fruiting stage, apply 500 g DAP + 100 g MOP per tree at the lemon size stage, and 200 g 19:19:19 + 100 g MOP at the guava size stage.

**INM in Kachai Lemon**

Imidacloprid on new flushes, followed by Copper Oxychloride (3g/l) after 15 days. Fertilize with a mixture of 650g Urea, 1220g SSP, and 450g MOP per tree. Apply a mixture of micronutrients (Zinc sulphate, Magnesium sulphate, Copper sulphate, and Manganese sulphate) at 0.25% on new flushes. In August-September, apply Bordeaux paste up to 60 cm on the trunk and repeat the spray schedule from June-July. For trunk borer grubs, clean the bored holes with iron wire, insert a petrol-soaked cotton swab, and seal with mud.

Banana Cultivation

Propagate with 4-month-old sword suckers for planting, with recommended characteristics such as weight between 1-2 kg and height between 80-120 cm. Prepare pits of 30 cm x 30 cm x 30 cm for planting, maintaining spacing from 2m x 2m to 3m x 3m based on soil fertility and management practices. At planting, apply 8-10 kg of FYM per plant and a fertilizer mixture of 180g N, 100g P₂O₅ and 250g K₂O per plant annually. Half the potassium and all the phosphorus should be applied during planting, with nitrogen applied in three splits over the following months. Remove daughter suckers until the mother plant flowers, and ensure proper propping with bamboo poles after flowering.

LIVESTOCK

Ensure optimal health and hygiene in livestock and poultry by adopting best practices in animal husbandry, including proper feeding, clean and dry housing, adequate ventilation, and access to clean water. Monitor livestock and poultry regularly, at least twice daily, for any signs of abnormal behaviour, posture, feed intake, rumination, or irregular urination and defecation. Implement strict biosecurity measures by limiting visitor access and preventing the introduction and spread of diseases on the farm. Dispose of carcasses responsibly using methods such as burning, deep burial, or designated disposal pits, in compliance with health regulations. Deworm animals and poultry routinely for both internal and external parasites, in consultation with a veterinarian, the State Animal Husbandry Department, or the Krishi Vigyan Kendra (KVK). Follow a comprehensive vaccination program for poultry (Ranikhet disease, IBD, Fowl Pox, and Marek's disease), pigs (Swine Fever), and cattle/buffaloes (FMD, Black Quarter, and Haemorrhagic Septicaemia).

FISHERIES

To ensure effective and healthy fish farming practices, proper pond preparation and management are crucial. Begin by applying lime (quick lime) at 200-300 kg/ha/year

if red diarrhoea symptoms are observed. Clean the poultry house regularly with phenol (1 ml/l of water) to prevent infections. Vaccinate birds against diseases like Ranikhet, Marek's, Gumboro, and Fowl pox as per the vaccination schedule. Report any unusual deaths or illness to the nearest veterinary center immediately. Avoid waterlogging near poultry houses to prevent flies and mosquitoes, ensuring a clean and dry environment. Turn litter materials weekly to maintain hygiene and reduce disease risks.

Fishery

Farmers are advised to carry out de-silting of ponds to facilitate further fish farming, followed by the application of lime and repairs to ponds, dykes, and inlet-outlet systems. Pre-stocking management for composite fish culture includes clearing aquatic weeds and regularly netting out predatory and weed fish to maintain a healthy ecosystem. Any surplus common carp brood fish should be disposed of accordingly. It is also the breeding season for exotic carps and Indian major carps, so breeders should begin production of fish fingerlings. Farmers are advised to avoid exceeding a stocking density of 10,000 fingerlings per hectare. Fish stocking should follow a ratio of 3:4:3 (surface feeder: marginal feeder: bottom feeder) for optimal growth. Additionally, fish growers can prepare fish feed at home by mixing mustard cake, rice polish, and cooked rice. For paddy-cum-fish farming, land preparation should begin to ensure proper conditions for both crops.

MUSHROOM

Growing oyster mushrooms (*Pleurotus spp.*), especially *Pleurotus djamor* (pink oyster mushroom), indoors using paddy straw can provide additional income for farmers and improve the nutritional status of their diet.

SELF-HELP GROUP/FPOs

Women members in SHGs should establish Nutrition Gardens in each household to provide fresh vegetables rich in vitamins and minerals, enhancing immunity. Vegetables such as brinjal, capsicum, chili, and cucurbitaceous vegetables can be grown in the kitchen garden. Additionally, SHG members can make masks, hand gloves, and cotton clothes for their family members and the neighbouring community, ensuring safety while generating additional income. FPOs should begin their business by producing high-quality seeds and planting materials through organic cultivation of ginger, turmeric, and pineapple, along with value addition.

proper drainage, and providing immune boosters with cool baths to mitigate heat stress. Regular monitoring and these integrated practices will enhance productivity while preventing disease outbreaks in your piggery operations.

Fisheries

Farmers are advised to prepare their ponds by maintaining a maximum depth of 1.75 meters, which is ideal for growing fry into healthy fingerlings (10-15 cm). Begin by applying an initial dose of cow dung manure at 10,000 kg/ha, followed by regular applications of urea and superphosphate (40-80 kg/ha every 15 days) to promote plankton growth, while maintaining optimal water quality through lime application at 250 kg/ha. For stocking, introduce fry at a density of 1 lakh/ha using recommended polyculture ratios of either 1:2:2 or 3:4:3 for Catla, Rohu, and Mrigal species, progressing to fingerling stocking at 5,000-8,000/ha in a 30:30:40 ratio as the fish mature. Farmers are advised to practice Happa breeding for common carp between June 1st and July 31st, and begin stocking of locally produced fingerlings from the first week of July to ensure better survival rates and disease-free stock. For feeding, provide commercial pellets at 2% of body weight, or alternatively, use a nutritionally balanced 1:1 mixture of rice bran with either mustard oil cake or palm oil cake when pellets are unavailable.

CALENDAR OF OPERATION FOR REJUVENATION OF KHASI MANDARIN

Trunk borer control - carefully clean infested holes with an iron wire before applying treatment either by inserting a cotton swab soaked in petrol or Dichlorvos, or by injecting 5 mL of 0.2% Dichlorvos solution (prepared at 2ml/L of water) and sealing the hole with mud to effectively eliminate grubs. As spring arrives, continue proper nutrient management by applying the annual dose of 30 kg FYM per tree along with balanced chemical fertilizers (650 g urea, 1220 g SSP, and 450 g MOP per tree). Implement comprehensive plant protection measures including: application of Bordeaux paste on tree trunks up to 60cm height; foliar spraying of Multiplex (2.5g/L) on new leaves; a protective spray of Bavistin (1g/L) combined with Monocrotophos (1mL/L) on new flushes; Micronutrient supplementation through foliar application of Zinc Sulphate (0.5%), Magnesium Sulphate (0.2%), Copper Sulphate (0.4%), and Manganese Sulphate (0.4%). For management of bark-eating caterpillars, inject 0.1% Dichlorvos directly into their tunnels. Complete these spring operations by mulching the basin area to conserve soil moisture and regulate soil temperature.

Khasi Mandarin

Locate the trunk borer and seal the holes with mud after inserting cotton soaked in petrol/kerosene. Shake the tree to manually kill the adult trunk borers. Install a light trap to capture adult trunk borers or pick them by hand. Use a spike to hook and kill the grubs of the trunk borer. Cut and burn the dried parts of the plants, as they harbor grubs and pupae. Generate smoke in the orchard for 2-3 hours after nightfall to control fruit-piercing moths. For controlling leaf miners, mix neem oil at a rate of 10 ml/liter of water and apply it during bud burst, repeating the application after 10 days if necessary. Prune the branches that are severely affected. To control bark-eating caterpillars, gummosis, and foot rot, keep trunks clean, apply Bordeaux paste up to 60 cm height in winter and early summer, and for caterpillars, insert cotton soaked in petrol or kerosene into tunnels and seal with mud.

For rejuvenation of Khasi Mandarin: In May, collect and destroy trunk borer adults by shaking the branches 2-3 times at 10-day intervals. Apply Carbaryl paste (20g/L of water) on the trunk up to 6 feet from the ground level. To prevent excessive fruit drop, spray NAA at 50 ppm (Planofix 0.5 ml/L). During June and July, spray a mixture of Zinc Sulphate (0.4%), Magnesium Sulphate (0.2%), Copper Sulphate (0.4%), and Manganese Sulphate (0.4%) on the new flushes. Apply fertilizers at the rate of 400g Urea, 1000g SSP, 400g MOP, 200g Zinc Sulphate, and 50g Borax per tree. After 15 days, spray Bordeaux mixture (1%) or Blitox-50 (3g/L). In August, apply Bordeaux paste on the tree trunk up to 60 cm height from the ground. Spray Copper Oxychloride (3g/L of water) and Dimethoate (1.5 ml/L of water). Drench the soil in the basin area with Mefenoxam, Fosetyl, or Ridomil (2.5g/L of water). For trunk borer management, clean the bore holes using an iron wire, insert a cotton swab soaked in Dichlorovos solution (2 ml/L of water) or inject 5 ml of the solution into the holes, and then seal with mud. Additionally, apply fertilizers at 400g Urea, 1000g SSP, and 400g MOP per tree. To prevent fruit fly infestation, spray Malathion (2 ml/L of water) over the fruits and install pheromone traps.

FRUIT CROPS

Banana

Select healthy suckers, dry them in direct sunlight for 3-4 days, dip in cow dung slurry and ash to prevent rhizome weevil, and periodically remove dried leaves to maintain clean cultivation. Prune the side suckers every month which attracts rhizome weevil and pseudo stem weevil. Proper disposal off the banana plant after harvesting and do not dump infested materials into manure pit. Apply neem cake @ 1 kg/plant to prevent rhizome and pseudo stem weevil infestations, plant until May, ensure proper drainage to avoid water stagnation, and prop banana plants with bamboo to prevent lodging due to heavy winds.

Kiwi

Provide protection to kiwi plants using 50% shade net for reducing flower and fruit drop due to hail storm. Regular weeding should be carried out, especially after early monsoon showers. Apply mulch (straw, dry leaves) around the base of kiwi plants to conserve soil moisture and suppress weed growth. Ensure proper drainage around plants to prevent waterlogging and fungal infections due to high humidity during the



Punjab's agriculture is characterized by the cultivation of wheat and rice, which together dominate over 80% of the state's total cropped area. Other significant crops include sugarcane, maize, oilseeds, cotton, and various fruits and vegetables. The state has a high crop intensity, with many fields being sown multiple times annually, contributing to its status as a major food producer in India. Punjab's agriculture faces several challenges, including monocropping, depletion of water resources, excessive use of pesticides and fertilizers, etc. These issues contribute to environmental degradation, economic stress for farmers, and ultimately impact crop production. There is utmost need to educate the farmers with agro-advisories regarding advanced techniques in cultivation of field crops, vegetable & fruit crops, agroforestry and livestock management in order to promote the scientific cultivation along with conservation of natural resources. The following agro-advisory highlights the important points to be followed by the farmers during kharif season to sustain the productivity of field crops, horticulture crops, agroforestry and livestock.

SOIL TESTING

Soil testing is the best tool to ensure optimum and balanced use of nutrients. Improper use of fertilizers leads not only to imbalanced nutrition but also deterioration of the environment. For collection of soil sample scrap away surface litter and make a V-shaped cut with a spade or a khurpa to a depth of 6 inches. Remove about 1" thick uniform slice of soil from one side of the cut. Similarly, collect samples from 7 to 8 places in the field of uniform texture and uniform fertility. Put the samples in a clean bucket, tray or cloth and mix it thoroughly. Take approximately half kg of soil in a cloth bag and label it with information such as field number, name of the farmer, address, date of sampling etc. If the samples are wet, dry them in shade before putting into the cloth bag. Soil and water samples are tested by Soil Testing Laboratory, Deptt. of Soil Science, Punjab Agricultural University, Ludhiana and Soil Testing Laboratory of all KVK's.

COMMERCIAL CROPS**Cotton**

Selection of varieties: To obtain higher yield, grow Bt cotton variety PAU Bt 2 and PAU Bt 3 and other Bt hybrids recommended by PAU, Ludhiana. Among the non Bt hybrids/varieties, grow F 2228, LH 2108 and desi cotton varieties LD 1019, LD 949 and FDK 124.

Sub Soiling: Cross sub soiling at 1.0 m spacing should be done before preparing the field with tractor drawn sub-soiler (chiseler) to the depth of 45-50 cm. This will help in breaking the hard pan, increasing water infiltration rate and better root development of cotton plants.

Time of Sowing: Sow the crop during 1 April to 15 May. Sowing during this period ensures better yield and escapes the attack of insect pests and diseases.

Pre-sowing irrigation: Apply heavy pre-sowing irrigation preferably with canal water for good germination and early crop establishment.

Seed rate: For Bt varieties PAU Bt 2 and PAU Bt 3, use seed rate @ 4.0 + 1.0 (refuge) kg/acre, for Non-Bt varieties F 2228 and LH 2108, 3.5 kg/acre and for recommended Bt hybrids, use two pouches of 475 g each (in which refuge Already mixed in seed)

Seed treatment: Soak the cotton seed in a solution of 0.5 g succinic acid in 5 litres of water for 2-4 hours for acid delinted seed or 6-8 hours for non-delinted seed. To prevent attack of jassid at early stage, smear the seed with 5 g Gaucho 70 WS (imidacloprid) or 7 g Cruiser 30FS (thiomethoxam) per kg seed. To remove the fuzz of non-delinted seed, rub these seeds with fine earth, cow-dung or ash.

Planting geometry: Sow the seeds with cotton sowing drill or cotton planter with line to line spacing of 67.5 cm. Maintain plant to plant distance of 30 cm for variety PAU Bt 2 and PAU Bt 3; 45 cm for desi cotton varieties; 60 cm for non-Bt hybrid/varieties and 75 cm for Bt hybrids by thinning out extra plants after first irrigation.

Weed management: Hoe the crop two or three times. The first hoeing should be done before first irrigation. For control of weeds particularly itsit, madhana/ makra, apply 1.0 litre per acre Stomp 30 EC (pendimethalin) as pre-emergence within 24 hours of sowing. In situations where weeds emerge after first irrigation or with the rain shower, Stomp 30 EC can also be applied as post-emergence after first irrigation in 200 litre of water. Alternatively, spray 500 ml per acre Hitweed Maxx 10 MEC (pyrithiobac sodium 6% quizalofop ethyl 4%) by dissolving in 150 litres of water after first irrigation, in moist soil, to control annual grass and broadleaf weeds. This herbicide also provides effective control of lapeta (guara) vel (Ipomoea sp.) when weed plants are at 2 to 5 leaf stage.

Fertilizer Application: Apply urea @162.5 kg to non-Bt varieties, 200 kg to Bt variety (PAU Bt 2 and PAU Bt 3) and 225 kg to all Bt and non-Bt hybrids, DAP @ 67.5 kg or SSP @187.5 kg per ha. Omit phosphorus (DAP or SSP) application to cotton when it follows wheat which had received recommended dose of phosphorus. Reduce the dose of urea by 25 kg when 67.5 kg DAP is applied to cotton. Apply 50 kg muriate of potash and 25 kg zinc sulphate heptahydrate (21%) or 16.25 kg zinc

sulphate monohydrate (33%) per ha to cotton in light soils. Apply 62.5 kg/ha of magnesium sulphate. Apply full dose of phosphorus, magnesium sulphate, muriate of potash and zinc sulphate at sowing. Apply urea in two splits at thinning and the appearance of flowers.

Foliar application of nutrients: To get higher yields, give 4 sprays of 2% potassium nitrate (13:0:45) at weekly interval starting at flower initiation.

Irrigation and Drainage: The first irrigation should be given 4 to 6 weeks after sowing and the subsequent ones at interval of two or three weeks. The crop must not be allowed to suffer from water stress during the flowering and fruiting stages. Cotton during its early growth is very sensitive to water stagnation.

Sugarcane

Sugarcane can be successfully grown on all types of soils ranging from sandy loam to clay loam and it is semi-tolerant to sodicity and salinity. Cross sub-soiling at 1.0 m spacing should be done once in three to four years, before preparing the field. This will help in breaking the hard pan, increase water infiltration rate and better penetration of roots.

Selection of varieties: Grow PAU recommended varieties i.e., early maturing-CoPb 95, CoPb 96, Co 15023, CoPb 92, Co 118, CoJ 85, CoJ 64, mid-late maturing- CoPb 98, CoPb 93, CoPb 94, Co 238, CoPb 91, CoJ 88.

Time of sowing: Mid-February to the end of March is the optimum time for planting sugarcane in the Punjab. Do not plant early maturing varieties after March. Do not plant the crop in a field where the preceding crop was infected with red rot or wilt.

Selection of seed and Seed rate: The seed should be free from red-rot, wilt, smut, ratoon-stunting and grassy shoot diseases. Use 20 thousand three-budded setts or 15 thousand four-budded sets or 12 thousand five-budded setts per acre. Use only the top two-third portion of the selected canes for planting.

Seed treatment: To improve the germination, soak the setts in ethrel solution overnight. To prepare the solution, dissolve 25 ml of Ethrel 39 SL in 100 liters of water. Alternatively, soak the setts in water for 24 hours before planting

Weed management: Two or three hoeings can be done with a triphali or tractor mounted tiller or tractor operated rotary weeder.

Chemical Control: Pre-emergence application of 800 g per acre Atrataf/Solaro/Masstaf/ Markazine 50 WP (atrazine) or Sencor 70 WP (metribuzin) or Karmex/Klass 80 WP (diuron) or 1000 g per acre Authority NXT 58 WP (sulfentrazone+clomazone) in 200 litres of water within 2-3 days of planting effectively controls the broadleaf weeds and annual grasses. For control of hardy weed like Bans Patta, use only Sencor 70 WP or Karmex/Klass 80 WP. Alternatively, post-emergence application of 1200 g per acre Triskele/Trishuk(2,4-D sodium salt 44% + metribuzin 35% + pyrazosulfuron ethyl 1.0%) WDG or 1000 g per acre Sindica (2,4- D sodium salt 48% + metribuzin 32% + chlorimuron ethyl 0.8%) WDG in 200 litres of water at 3-5 leaf stage of weeds provides effective control of seasonal grass, sedge and broad leaf weeds. For fields infested with dila, post-emergence application of 800 g per acre 2, 4-D sodium salt 80 WP in 200 litres of water is recommended. In

25 WG (thiamethoxam) or 6.25 litre MAK HMO (Horticulture Mineral Oil) in 500 litres of water per acre in the first week of September. Even spray of Soybean oil @ 10 litres or 10 litres of cotton seed oil + 1.25 kg of detergent powder in 500 litres of water is equally effective. There should not be any water stress in the orchard at the time of spray of MAK HMO, soybean oil and cotton seed oil. During extreme hot months spray should be done in the evening hours. To manage fruit flies, regularly collect and bury the infested fruits. Fix PAU fruit fly traps @16 traps/ acre in the second week of August and recharge the same if required. Spray soybean oil @ 10 litres or 10 litres of cotton seed oil + 1.25 kg of detergent powder in 500 litres of water per acre in May-June and August-September as soon as mite population is observed on the underside of leaves. To manage Foot Rot/ Gummosis, avoid flood irrigation. Do not allow irrigation water to stay in direct contact with the tree trunk. Avoid deep hoeing. Avoid injuries to the trunk bark and roots. Give two applications of Curzate M 8 as paint (2 g/100 ml of Linseed oil) to the infected trunk portion and drench (25 g/10 litres of water/ tree) the soil at the base of the tree in July-August. During last week of June to August, apply three sprays of Bordeaux mixture (2:2:250) or 50 % Copper oxychloride (3 g per litre of water) at 20 days interval to manage scab which appears as small, dark brown, rough, irregular, raised lesions mostly on underside of the leaves, do clipping of infected leaves in July Give light and frequent irrigations in May-June and avoid water stress to combat problem of fruit cracking.

Guava

Apply Farmyard manure in May. Application of 20 kg Paddy straw manure can be used as a substitute of Farm Yard Manure to full grown guava plants. To full grown guava trees, apply urea @ 1.0 kg/plant, SSP @ 2.5 kg/plant and MOP @ 1.5 kg/plant. Half of the inorganic fertilizers should be applied in May-June and the remaining half in September-October. Correct Zinc deficiency by spraying guava trees with zinc sulphate (1 kg of zinc sulphate + 1/2 kg of unslaked lime in 100 litres of water). Give 2 or 3 such sprays at fortnightly interval between June to September. Typical symptoms of Zinc deficiency are narrowing of leaves; interveinal chlorosis and twig die back. For weed management, apply paddy straw @ 4.0 t/acre as mulch after application of recommended dose of organic manure and inorganic fertilizers in May under the tree canopy. Guava gives two crops in a year. Winter season crop is superior in quality than rainy season crop, as rainy crop may get infested with fruit flies. Those farmers who intend to get only winter season crop, can spray urea 10 per cent (100 g



Fruit Fly in Guava



Anthrachnose in Guava

Rajasthan is predominantly a kharif cropping state due to vast area under rainfed cropping and crops. The diverse crops are grown during Kharif season. During May and June preparations for sowing of different crops like maize, bajra, green gram, cowpea, guar, moth bean are completed by the farmers. The sowing and nursery raising of cotton, groundnut and rice are done under varying situations and climatic conditions. To harness the potential of the crops, an appropriate technology along with management practices are inevitable. Therefore, a proper agro-advisory is need of the day to be issued for the use of the farmers for different agricultural operations. It will help in increasing the productivity and profit from agriculture and empower the farmers while helping in sustainable management of natural resources and applied inputs. The agro-advisories related to different activities and operations in agriculture conducted by the farmers during May-June are discussed in following para:

Soil Analysis

After the harvesting of rabi season crops in March and April and before sowing of kharif crops is the best time for taking soil samples from the farmers' fields for their analysis for physical, chemical and biological properties. The farmers are advised to take soil samples from 5 places of their fields, mix it properly and take them to nearby KVK or district soil testing labs for analysis. The sampling should be done from 0-30 cm soil depth. Attempt should be made get the soil sample analysed before sowing of kharif crops. The nutrient status as evidenced from soil sample analysis will help to decide right and balanced application of fertilizers and avoid extra application. It will also help in enhancing the nutrient use efficiency and saving of cost of cultivation while maximising the profit to farmers.

Summer Deep Ploughing and Application of Gypsum

May and June are the best time for summer deep ploughing and application of Gypsum in the field. The summer deep ploughing is process of solarization of the soil

Seed Treatment: For management of seed borne diseases treat seed with Thiram 2 g/kg + Carbendazim 1 g/kg or *Trichoderma viride* 5 g/kg seed. For bacterial blight management soak the seed for 30 minutes in 2 gm solution of *Streptomycin* in 10 litre of water prior to seeding. By treating seed with *Azotobacter* and *PSB* fertilizer 25% nitrogen and phosphorus can be saved.

Improved Varieties: RT- 392, RT- 372, RT-346, RT-46, RT-125, RT-127, RT-103, RT-35, TC-25 (spreading type), Pratap (C-50), etc.

Seed Rate: For branching types, the seed rate is 2-2.5 kg per ha with plant geometry of 30-35 cm x 15cm. For non-branching types of varieties (Pratap) the seed rate 4-5 kg per ha with plant geometry of 30 cm x 10-15 cm.

Nutrient Management: Farmyard manure or compost is applied @ 20-25 tons per ha 15-20 days before sowing. In sufficient rainfall areas, 25 kg phosphorus and 20 kg nitrogen should be given in furrows at the time of sowing 4-5 cm below the seed. After 4 to 5 weeks, the remaining nitrogen sprinkled during light rain. In areas with less rainfall, reduce the amount of fertilizer and use phosphorus based on soil test. In zinc deficient fields, spray 0.5 percent zinc sulphate plus 0.25% lime solution at 50 days after sowing. Use of gypsum 150 kg/ha can increase the yield as well as oil content in sesame.

Water Management: Sesame is very susceptible to drought at various physiological growth stages. The crop requires about 50 cm of water during the entire growth period. Usually, no need to irrigation is required if rains are well distributed. However, in case of long dry intermittent spells, adequate moisture in soil should be maintained with light irrigations.

Weed Management: First weeding at 20-25 DAS & Second weeding at 30-45 DAS. Fluchloralin (Basalin) 1 kg a.i. /ha in 800-1000 liters water as pre-planting effectively manage weeds.

Plant Protection: *Phyllody*: two time spraying of Dimethoate 25 EC @ 750 ml or Quinalphos 25 EC @ 1 litre per ha, first at 25-30 DAS or second at 40-45 DAS. Alternatively spray 5 % neem leaf solution twice @ 21 & 35 DAS. Leaf & pod borer: spray quinalphos 25 EC @ 1-1.5 litre per ha in 500-700 liters of water and repeat at an interval of 15 days as per requirement. Hawk moth and Gall fly: Apply Quinalphos (1.5%) powder @ 25 kg per ha. Blight: spraying mancozeb @ 1.5 kg or captan @ 2 kg per ha at the interval of 15 days.

COMMERCIAL CROPS

Cotton

Sowing: The cotton performs well when sown between first week of May to third week of May. The best time of its sowing is before 15 May.

Seed Rate and Spacing: *Desi (arboreum)*: 12-15 kg/ha; (120 x 60 cm), *American (Narma)*: 8.75-10 kg/ha; (90 x 45 cm), *Hybrid Bt cotton*: 2.4-3.0 kg/ha (75 x 30 cm). These spacings result a plant population of ~40,000-50,000 plants/ha. The seed should be sown at 2.5-3 cm depth and maintain 2-3 plants at point after thinning.

Seed treatment: Delinting: Dip seeds in concentrated sulfuric acid (1 L/10 kg seed) for 1-2 minutes until black. Rinse thoroughly and discard floating seeds. Dry in shade. (Use plastic containers only; wear gloves and dispose of acid safely.)

Pre-sowing Treatment: Soak dried seeds in water for 6-8 hours. Treat with fungicide (e.g., carbendazim 2 g/kg or *Trichoderma* 6-8 g/kg seed). Apply systemic insecticide (e.g., imidacloprid 70% WG at 5-7.5 g/kg seed) for early pest control.

Fertilizer Application: Incorporate 8-10 tonnes/ha well-decomposed FYM in soil thoroughly before sowing. Apply chemical fertilizers based on soil test recommendations. If no soil test is available, apply 90-100 kg nitrogen/ha for desi cotton and 100-150 kg nitrogen/ha for American cotton, divided into 2-3 split doses. Also apply 20-40 kg P_2O_5 /ha at sowing. In case of Zinc deficiency, apply zinc sulfate at 25 kg/ha, or spray borax 0.1% during flowering if boron is deficient in soil.

Improved Varieties: *Desi (arboreum)*: HD 123, RG 8, RG 542, CICR-1, CICR-3, KR 64. *Narma (American)*: RS 875, RS 810, RS 2814, RS 2013, Bikaneri Narma. *Bt Hybrids*: RAJ H-HH-16, CSH 198, CSH 238. Buy certified seeds only and follow refuge planting for Bt cotton.

Water Management: Light irrigation within 3-4 days of sowing if no rainfall. First full irrigation can be given around 20-30 days after sowing. In water-scarce areas, use drip irrigation and mulching to retain moisture.

Plant Protection: To control leaf sucking pest & Leaf Curl Virus disease, seed should be treated with 4 g Thiamethoxam 70 WS/kg seed before sowing. Keep close monitoring on White fly host plants like vegetables, flower plants, weeds & unwanted plants and from time to time these affected host plants must be uprooted & burn. Some vegetables like Ladies Finger, Brinjal, Tomato & Chilly also act as host plants for white fly, accordingly, as per need these must be sprayed with Thiamethoxam 25 WG @ 0.5g/litre water. To control bacterial blight seed should be treated with soaking in 1 g *Streptomycin* or 10 g *Plantomycin*/10 litre water before sowing. Spray Thiamethoxam 25 WG @ 100g/ha and Profenophos 50 EC @1250 ml/ha for sucking pest. Proclaim (emamectin benzoate) 5 SG @250g/ha for pink, spotted and American bollworm

FRUIT CROPS

Pomegranate

Climate and Soil: Dry and semi-dry climate is very suitable for pomegranate production. Sandy loam soil with good drainage is the most suitable. Pomegranate plants have good capacity to tolerate salt and alkalinity.

Improved Varieties: 1. Bhagwa, 2. Sonar Lal, 3. Super Bhagwa

Planting Material: Tissue culture saplings are the best to guarantee disease-free planting material and for planting in new areas where there are no pomegranate orchards.

Establishment of Orchard: The digging of pits should be completed in May. Square and rectangular method is the easiest and convenient for planting orchards. The size of pit should be 60 x 60 x 60 cm at 4 meters rows apart. The plant to plant spacing should be 3 metres.

Planting Time: July-August is the suitable time for planting saplings. The saplings should be planted in the evening. The soil should be pressed well around the sapling after planting. The plant should be irrigated immediately after the plantation.

of Chloropyriphos 20 EC (1 ml/L) help to manage it, destroy alternate host plants growing nearby. Shoot Borer: Prune and burn affected branches. Spray branches and trunks with a solution of Chloropyriphos of 2 ml/ L of water. Clean the cavities with a pesticide-coated stick or cotton, then seal with mud and cow dung paste. Root knot nematode: To manage practice soil solarization, use healthy planting material, apply organic manures, inter crop marigold plants, use resistant varieties and apply 5-10 kg neem cake per plant to suppress nematodes. Wilt Disease: control is difficult but drenching the soil around the root zone with 20-30 liters water per plant of carbendazim 50 wp 2 gram/L fungicide solution can help. Mix 1 kg of *Trichoderma viride* per 10 q of compost or farmyard manure and apply around the root zone, after keeping it 15-20 days under shade. Use 1-2 kg *Trichoderma* mixed compost per plant. *Anthracnose*: To manage this remove and destroy the infected twigs and plant parts. Spray Mancozeb 2 gm per liter of water every 10-15 days until fruiting begins. Fruit Rot: Spray Mancozeb or Zineb 2 gm per liter of water every 15 days at least twice. For already infected plants, spray preventively before rainfall.

Harvesting and Yield: For local markets, harvest when fruits are fully ripe. For long-distance market fruit should be harvested when mature but still firm, when fruits change from green to reddish in 24 hours or more, harvest them promptly. A single plant can produce around 40-50 kg of fruits, and mature trees (8-10 years) can yield up to 100 kg per tree. Good practice of grading and sorting, packing, transport and marketing helps to fetch better prices.

VEGETABLE CROPS

Brinjal

Improved Varieties: Pusa Purple Long, Punjab Sadabahar, ARU-2, H-4, H-7, PB-91-2, K-202-9, DBR-8, AB-1, Hybrid varieties- Arka Navneet, Pusa Hybrid-6, Pusa Hybrid-2

Nursery: 400-500 grams of seeds are required for one hectare. For one hectare of seedlings, 15-20 beds of 1 meter width and 3 meters length and 10-15 cm height are required. Seeds should be treated with thiram or captan 2 g/kg. In the nursery, seeds should be sown in rows at an interval of 5 cm.

Field Preparation: The field should be ploughed 3-4 times deep. First ploughing should be done by mixing 120-150 quintals of decomposed cow dung manure/hectare in the soil. During the last ploughing, 40 kg nitrogen, 80 kg phosphorus and 60 kg potash/hectare should be mixed in the soil.

Sowing Time: Nursery Preparation: June-July; Transplanting in the main field: July-August

Planting: Saplings should be transplanted in the evening when they are 10-15 cm and 30-40 days old. Distance between rows should be 60-70 cm and distance between plants should be 60 cm.

Manure and Fertilizer: 20 kg/hectare nitrogen should be sprayed when flowers appear.

Irrigation: Irrigation should be done every 4-5 days or as per requirement in rainy season.

Stage-Specific Inoculants: Biofertilizer pack with *Azospirillum*, *Azotobacter*, *Phosphobacteria*, *Potash bacteria*, *Pseudomonas*, and PPFM for different crop stages; enhances yield by 10-15% and reduces chemical fertilizer use by 25%. **Zinc Solubilizing Bacterial Biofertilizer:** Carrier-based or liquid form; enhances Zn nutrition and yields by 17-20%, adding 250-400 kg/ha. **TNAU Micronutrient Mixtures:** Enriched FYM-based micronutrient mixtures for wetland and rainfed rice; alleviates deficiencies, improves quality, and enhances yield by 11-20%. **Nanosci Lure for YSB:** Nano-pheromone lure for eco-friendly pest monitoring and mass trapping of yellow stem borer. **Stage-Specific Inoculants:** Biofertilizer pack tailored to growth stages; increases yield (10-15%) and reduces chemical fertilizers by 25%. *Bacillus altitudinis* FD 48: Dual-use bioagent (seed+foliar) enhancing root traits and yields under both irrigated and stress conditions. **TNAU Biomineralizer for Stubble:** Microbial product (2 kg/tonne) for in-situ rice straw decomposition to reduce burning and enrich soil.

Maize

Nutrient & Soil Management: Neem cake @ 250 kg/ha; Zn citrate (0.5%) + Fe citrate (1%) foliar sprays at 30, 40, and 50 DAS in deficient soils. **FAW Management (Technology Capsule):** Seed treatment: Thiamethoxam 30 FS @ 10 ml/kg; maintain spacing (60 × 25 cm) + rogue rows every 10 rows. **Border cropping:** Cowpea, gingelly; **Intercrop** with blackgram; install pheromone traps (50/ha) and solar traps (1/ha). **Insecticides by stage:** Early whorl (15-20 DAS): Emamectin benzoate 5 SG, Chlorantraniliprole 18.5 SC, Flubendiamide 480 SC, or Azadirachtin. Late whorl (35-40 DAS): Metarhizium anisopliae, Emamectin benzoate, Novaluron, or Spinetoram. **Tasseling/cob formation** (60-65 DAS): Use any insecticide not used earlier. **Biologicals:** Release *Telenomus remus* @ 1.25 lakh/ha.

COMMERCIAL CROPS

Cotton

High Density Planting: Closer spacing (90 × 15 cm) for 29,629 plants/acre using suitable varieties like Suraksha, CO-17, and VPT-2.

Fertilizer: Apply 100:50:50 kg NPK/ha; weed control using Pendimethalin 38.7% CS @ 700 ml/acre within 24-48 hours of sowing.

Growth Management: Spray NAA @ 40 ppm or Cotton Plus @ 2.5 kg/acre; apply Mepiquat Chloride (0.015%) at square/flowering stage; use sodium chlorate (0.9%) or Drop Ultra as defoliant at 60% boll burst.

Intercropping: Paired row planting (45/90 × 30 cm) of Bt Cotton with two rows of cluster bean; application of Mepiquat Chloride and Sodium Chlorate enhances suitability of CO 17 for mechanical harvesting.

Coconut

Pest Traps: Install yellow sticky traps (20/ha or 8/ac), trunk banding, and spray jet water on fronds. **Biocontrol:** Release *Apertochrysa astur* (1000/ha), *Encarsia spp.* (250/ha or 100/ac). **IDM Package:** FYM (50 kg) + neem cake (5 kg) + fertilizers; CuSO_4 (200 g) + MgSO_4 (1 kg) alternately; bio-agents (*Trichoderma*, *Bacillus*, *Phosphobacteria*, *Azospirillum*, VAM); TNAU Coconut Tonic (40 ml twice/year);

Micronutrient Spray (Tomato, Bhendi, Chilli): TNAU multi micronutrient @ 1% during vegetative and flowering stage. Bhendi: Spray WSF 19:19:19 @ 2% + liquid Multi MN @ 1% at 30 DAS, repeated 3 times every 10 days.

Finger Millet

Organic Package: FYM @ 3 t/ha, Azophos (2 kg/ha), *Trichoderma* (2.5 kg/ha), vermicompost (1.5 t/ha), Panchagavya (3%) at vegetative, flowering, and grain filling stages. Disease/pest control: NSKE 5% and *Bacillus subtilis* 0.5% as needed. Foliar spray: 1.5% foliar formulation at flower initiation. Horsegram: Foliar Spray: TNAU Horsegram Wonder @ 5 kg/ha during flowering with wetting agent.

PULSE CROPS

Greengram, Blackgram, Redgram

Foliar Spray: TNAU Liquid Pulse Wonder @ 5 kg/ha at peak flowering (Greengram/Blackgram) and 15 days after (Redgram). Land preparation: Raised beds (90 cm) with 30 cm furrows. Seed Treatment: Imidacloprid (5 ml/kg) + *Bacillus subtilis* (10 g/kg) + *Rhizobium* & *Phosphobacteria* (30 g/kg). Agronomy: Spacing 30 × 15 cm; drip irrigation; fertigation 25:50:25:40 NPKS kg/ha. for weed control: Pendimethalin + Imazethapyr (PE) + Quizalofop-ethyl + Imazethapyr (EPoE). TNAU Vigour Plus: Nano-formulated seed coating (20-25 ml/kg) that boosts germination, seedling vigor, and early establishment. Bio-Pulse: Microbial seed bio-priming formulation that enhances yield under pathogenic stress, effective in chickpea, pea and lentil.

Blackgram

Nodule-Associated Plant Probiotics for Blackgram: Blend of beneficial microbes improving nodulation, drought tolerance, and yield (14% over standard). Newer fungicides for foliar disease management: Foliar spray of tebuconazole + trifloxystrobin (1g/L) or tebuconazole 25 EC (1ml/L), at symptom appearance and 15 days later; effective against *powdery mildew* and *Cercospora leafspot*.

Greengram

Newer fungicides for foliar disease management: Foliar spray of tebuconazole + trifloxystrobin (1g/l) or tebuconazole 25 EC (1ml/l), at symptom appearance and 15 days later; effective against powdery mildew and *Cercospora leaf spot*. TNAU Vigour Plus (Seed Booster): Pre-sowing seed coating with biopolymer matrix containing IAA and GA³; enhances seed germination and seedling vigour (also helpful in multiple crops like Pulses, Groundnut, Cotton, Paddy, Millets, Vegetables).

OILSEED CROPS

Groundnut

TNAU Groundnut Rich: A foliar nutrition solution. Applied twice @ 5.0 kg/ha at 35 DAS (50% flowering) and 45 DAS (pod development). Requires 500 litres of water per spray. Enhances pod development and improves yield. Nut Boost: A seed inoculation formulation containing a consortium of phosphate, potash, and zinc solubilizing PGPRs. Applied as a seed treatment. Improves pod yield by 16.5% to 18.1% compared to untreated seeds.



Farmers are advised to follow the sowing rule to take-up sowing of rainfed crops like Cotton, Soybean, Maize, Jowar, Redgram, Greengram etc., only after receipt of cumulative rainfall of 50-60 mm in light soils and 60-75 mm in heavy soils or the soil should be wet up to top 15-20 cm depth after enter of South-West monsoon rains into the state.

CEREAL CROPS

Paddy

Under assured irrigation farmers are advised to raise rice nurseries of long duration (140-145 days) varieties from May 20th to June 20th. Under assured irrigation farmers are advised to raise rice nurseries of medium duration (130-135 days) or short duration (120-125 days) varieties from June 20th onwards. Month of June is not optimum for raising paddy nurseries using Telangana Sona (RNR15048) variety. Take the Greengram as catch crop to preceding rice in areas where the release of water into the canals and tanks is delayed. Green manuring with Sesbania/ Crotalaria / Pillipesara / grain legumes, residues of blackgram / greengram and in situ incorporation before flowering. Varieties: cultivation of long duration (BPT-5204, WGL-44, RNR 11718 etc.), medium duration (WGL-14, JGL-11470, JGL-384, RNR-28361, JGL-33124 etc), short duration (RNR-15048, KNM-1638, KPS-6251, WGL-362, KNM-118, JGL-24423, RNR-29325, JGL-28639 etc.) varieties.

Direct seeding of rice with Seed-cum ferti drill wherever suitable. In Telangana state, dry converted wet and wet direct seeded rice are popular, and the sowing can be started from June ending. Treat seeds with carbendazim @ 3 g/ kg (dry seed treatment) or 1 g/L (wet seed treatment). Farmers are advised to apply the Carbofuran 3CG granules @ 800 g per acre of nursery one week before pulling of the nursery. Root dipping: prepare the slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000 g/ha) of phosphobacteria or 10 packets of (2000 g/ha) of Azophos, inoculant in 40 lit. of water and dip. the root portion of the seedlings for 15-

30 minutes in bacterial suspension and transplant. Make 20 cm alleyways at every 2 meters to facilitate free aeration, ease of farm operations and better pest and disease management. Broadcast butachlor @ 1-1.5 l or pretilachlor @ 500-600 ml or 4 kg granules of bensulfuron methyl (0.6%) + pretilachlor (6.0%) with 20 kg sand per acre. Practice alternate wetting and drying in well irrigated areas to save water, better aeration and minimize incidence of BPH besides getting carbon credits with less emission of greenhouse gases. Recommended dose of fertilizers: 40-48 Kg Nitrogen, 20 Kg Phosphorus 16 Kg Potash per acre during Vanakalam. Spray 0.2% ZnSO₄ 2 to 3 times at 5 days interval to manage Zn deficiency, particularly in alkaline soils.

Maize

Deep summer plough to destroy pests and conserve moisture. Optimum sowing window is from June 15th to July 15th with medium and long duration varieties (90-100 days and 100-120 days) without any yield reduction. Sow medium duration varieties/hybrids DHM-206, DHM-182, DHM-117, Bisco-855, Bio-9637, Bisco-740, Kaveri Ekka-2288, Kaveri -2020, Kaveri-3696 and short duration hybrids DHM-115, Pioneer-3342, KH-5991, DKC-7074 during 15th June to 15th July. Sowing can be extended up to July end if monsoon is delayed. Sow on receipt of at-least 50-60 mm rainfall for better germination and complete sowing in a week for reducing the incidence of FAW. Spray NSKE 5% or neem oil 1500 ppm @5ml/l after sowing and at weekly intervals. Dibble seeds with tractor drawn cultivator or seed cum ferti drill on ridges and beds. Intercrop red gram varieties PRG-176, WRG-97 in 2:1 ratio in light soils and in 4:1 ratio in heavy soils to reduce incidence of FAW. Follow IPM for FAW management: summer ploughing, growing single cross hybrids, clean cultivation, intercropping with pulses, balanced fertilization, release of egg parasitoids (*Trichogramma* sp) and spraying Chlorantraniliprole @ 0.4 ml or spinetoram @ 0.5 ml or Emamectin benzoate @ 0.4 g per litre in plant whorls. As post emergence, spraying of Tembotrione @ 115 ml + Atrazine 400 g or Toperamezone @ 40 ml + Atrazine 400 g or mesotrione + atrazine @1400 ml/acre in 200 litre of water will effectively control weeds. After 30-35 days, crop may be inter-cultivated followed by earthing up is needed for better nutrient uptake and to avoid lodging.

COMMERCIAL CROPS

Cotton

Sow the crop early by June 1st/2nd week after receipt of sufficient good rains at least (60mm). Don't sow cotton crop after July 20th as yields will be reduced drastically under rainfed conditions. Sowing of Cotton on ridge and furrow in medium soil and broad Bed furrow in black soil under rainfed conditions facilitates the crop to moisture conservation and excess water drain out. In a village or area all the cotton cultivating farmers should take up the sowings on community basis and complete within a period of 7 - 10 days. Depending on soil type and rainfall pattern cotton should be sown at a spacing of 120 x 60 cm/ 120 x 45 cm/ 90 x 60 cm/ 90 x 45 cm/ 90 x 30 cm. Deep summer plough to destroy the eggs of insect pests and to remove weed stock. Intercrop with greengram/ blackgram/ soybean/ cluster bean in 1:2 or 1:3 ratio, and red gram 4:1 / 6:1 / 8:1 ratio which also helps to manage sucking pests.

Grow 2-3 rows of maize or sorghum as border crop as barrier to prevent pests. Treat the seed with imidacloprid 70 WS@ 5 g per kg or thiamethoxam 70% WS @ 4 g per kg followed by *Trichoderma viridi* or *Pseudomonas fluorescens* @ 10 g per kg.

Pre-emergence herbicides like pendimethalin (Stamp extra) @ 700 ml/acre should be applied within 24 -48 hours after sowing with the help of hand sprayer for better control of weeds in cotton. Application of post emergence herbicides like quizalofop ethyl @ 400 ml + pyrithiobac sodium @ 250 ml in 200 L of water at 30 DAS in cotton to control grasses and broad-leaved weeds. Inspect the crop at squaring and flowering stage for presence of PBW larvae within flowers. Installation of Pheromone traps @ 8/acre and monitoring ETL at this stage is 10 % damaged flowers. Spray $MgSO_4$ @ 10 g/l twice at 45 and 75 DAS; Boron @ 1.5 g/l twice at 60 and 90 DAS; 0.5% $FeSO_4$ with citric acid twice at weekly intervals and $ZnSO_4$ @ 2 g/l twice at 4-5 days interval at 45 DAS to manage micronutrient deficiencies. During excess rainfall, drain excess water and apply 25 kg urea and 20 kg potash per acre. Spray Planofix (NAA 10 PPM) @ 1 ml per 4.5 liters of water twice at 10-15 days interval to manage flower dropping. Spray 1% KNO_3 or 19:19:19, 20:20:20, 21:21:21. Follow rotation of insecticide sprays against insect pests for high efficiency. Stem application of fipronil 1: 4 at 30 DAS and Flonicamid / Imidacloprid with water (1:20) at 45, 60 DAS to be done which protects the crop from early season sucking pests like jassids and aphids.

PULSE CROPS

Pigeon pea

Sow high-yielding wilt-resistant varieties (155-165 days) like WRGE-93, WRGE-97, WRGE-121, and medium-duration varieties like WRG-255 and TDRG-4, TDRG-59 & TDRG-272 can be recommended. Sowing should be initiated only after receipt of 60 mm rainfall. The optimum sowing time should be 15th June to 15th July at a spacing of 150-180 x 20 cm in black soils and 90-120 x 20 cm in red soils. Seed rate should be 2.5 to 3.0 kg per acre only (plant-to-plant spacing should be mandatory). Treat seeds with thiram or captan @ 3g/kg followed by rhizobium culture @200-400 g/ 3 kgs of seed. The raised bed method or ridge and furrow method should be followed to protect the crop from *Phytophthora* blight disease, which occurs due to excess and stagnation of rainwater. Intercropping in Red gram + Cotton (1:4/ 1:6), Soybean (1:4/ 1:6), Green gram/ Black gram (1:7) should be followed. Treat the Red gram seed with *Trichoderma viride* @ 10 g/ kg seed to protect the crop from soil-borne diseases.

Spray pendimethalin @ 1 - 1.5 L in 200 liters of water within 24 - 48 hours after sowing will effectively control the weeds in Red gram Post-emergence application of quizalofop ethyl @ 400 ml/ acre to control grasses and imazethapyr 10 % @ 300 ml/ acre to control grasses and broad-leaved weeds. Both herbicides in the tank mix are also recommended in case both weeds are observed. If there is submergence due to excess rain, drain excess water at the earliest and spray 1% KNO_3 or 19:19:19, 20:20:20 or 21:21:21. Application of Multi K or 13-0-25 @ 10g/l at the time of flowering should be done. Practice nipping at 45 and 60 DAS, wherein terminal branches are removed to facilitate lateral branches, thus improving yields. To control pod fly spraying of Thiomethaxam 25 wg @ 0.2ml/l, Flubendamide 0.2ml/L with Jaggery 5g/L to be sprayed at the time of early pod formation stage.

Greengram and Blackgram

Sowing time is generally from June 15th fortnight- July 15th during Kharif, Sept 15th to 30th October during Rabi, January last week to March 15th during Summer. Greengram varieties MGG-295, MGG-385, MGG-351, TM-96-2, WGG-37 and WGG-42, Virat, IPM-2-14, Sikha. Blackgram YMV-resistant varieties MBG-1070, PU-31, GBG-1 and TBG-104 are the preferred varieties. Treat seeds with imidacloprid @ 5g or thiamethoxam @ 5g/kg. Intercropping can be done with cotton (2:1 or 1:1 ratio) or redgram (6:1)/(8:1). Greengram can be taken as a preceding crop to Rice and Chilli as green manure for incorporation in the soil.

OILSEED CROPS**Groundnut**

Seed rate recommended is 237 kg/ha and practice seed treatment with imidacloprid 600 FS @ 2 ml/kg followed by tebuconazole 2 DS @ 1g or Trichoderma viride @ 10g/kg to control early and late leaf spot diseases, sucking pests, collar rot, PSND in the early stages. Application of Gypsum @ 500 kg/ha at pegging stage near podding zone for pod development. Weed management with Diclosulam 84% WDG @31 gm/ha as pre-emergence herbicide followed by post-emergence application of Imazethapyr @ 750 ml/ha at 20 to 25 days after sowing.

Soybean

Sow varieties ASB-22, JS-335, AISb-1, JS-93-05 during 15th June 1st week of July using seed drill/ gorru/ desi plough/ by tractor drawn seed cum fertilizer drill / broad bed furrow planter. Sowing of Soybean by Broad bed Furrow (BBF) planter or Raised bed method facilitates the crop to withstand heavy rains and moisture stress conditions and to obtain higher yields. Treat the seeds with carbendazim @ 1g or thiram or captan @ 3 g/kg followed by 1.5 ml of imidacloprid (48 %) and Rhizobium culture @ 200 g/10 kg seed. Apply 50 kg DAP + 26 kg MOP at the time of sowing and top dressing of urea @ 25 -30 kg/acre at 30 DAS. Pre-emergence herbicides like pendimethalin 30 % @ 1 - 1.5 L/ acre should be applied within 24-48 hours after sowing with the help of hand sprayer for better control of weeds. Post emergence application of quizalofop ethyl @ 400 ml/ acre for control of grasses and imazethapyr + imazomax 40 g /acre or imazethapyr 10 % @ 250 ml/acre to control the grasses and broad-leaved weeds. To control stem fly and stem gridler as a prophylactic application of carbofuran 3G granules @ 8 kg/acre is recommended at 10 DAS. To control of pod blight spraying of tebuconazole + sulphur @ 2.5 g/L is recommended. Dry the seed and maintain the below 12 % moisture during seed storage to avoid incidence of pest and disease.

Castor

Sow high yielding castor hybrids viz., ICH-5, ICH-66, PCH-111, DCH-519 & ICH-6, PCS-262 from 15th June to 15th July in rainy season under rainfed conditions. Intercrop with redgram in 1:1 or greengram in 1:3 ratio, open furrows between rows to conserve moisture in light soils.

LIVESTOCK**Animal Husbandry & Fisheries**

Agriculture and allied sectors play a vital role in the lives and livelihoods of the majority of Telangana's population. The state is rich in livestock resources and

liquids & probiotics in drinking water during summer. Utilize broken rice or bajra (up to 30-40%), cotton seed meal (10-12%) and rapeseed meal (5 -8%) in poultry when there is shortage of maize during natural calamities. Keeping in view of disease outbreak follow biosecurity measures and mass vaccinate backyard poultry birds against Newcastle and viral diseases to prevent outbreak during stress periods. For hatching purpose of fertile eggs keep it in fridge at temperature of 4°C. To improve the growth rate 1 per cent protein level (soybean meal) may be added in the feed. Suitable coccidiostat should be added in the feed continuously by consulting the local veterinarian to prevent coccidiosis. 3-5 g of oyster shell/limestone/grit per day/bird shall be given to laying hens to avoid leathery eggs.

Value Addition: Convert milk into ghee, paneer, curd, khoa for better returns. Explore meat processing and packaging for hygienic sale. Utilize manure for biogas and compost.

Fisheries

Failed Monsoon: Use bore well or alternative water sources to maintain minimum pond water levels. Reduce stocking density to lower water demand. Adopt partial harvest strategies to minimize losses. Use water-saving practices like reduced water exchange. Avoid new stocking if water availability is uncertain.

Flood: Strengthen pond bunds and spillways to prevent overflow and escape of fish. Install protective netting around pond perimeters. Avoid feeding during peak flood to prevent water quality issues. Harvest fish early if flood risk is imminent. Apply disinfectants like Benzalkonium Chloride 80% (BKC) @ 2.5 L/ha and Rock Salt @ 75-125 kg/ha/5ft depth/ha after floodwater recedes to control disease outbreaks.

Drought: Reduce biomass load in ponds by early or partial harvesting. Avoid use of organic manure if water level is too low to prevent oxygen depletion. Apply aeration measures (e.g., paddle wheel aerators) to maintain DO levels. Maintain good water quality by regular monitoring and minimum exchange.

Delayed Monsoon: Postpone seed stocking until adequate water level is available. Increase application of manures/fertilizers once rains begin to promote plankton growth. Avoid breeding activities during uncertainty to protect brood stock. Maintain brood stock in protected tanks.

Advanced Monsoon: Prepare ponds early for stocking, complete liming 200 kg/ha and fertilization in advance. Ensure brood stock is not disturbed during monsoon (15th June to 31st August). Avoid overfeeding as heavy rains can cause feed waste and water quality degradation. Watch for sudden temperature or pH fluctuations due to early rains.

High Temperature: Maintain higher water levels (5-6 feet) to buffer temperature fluctuations. Avoid feeding during peak heat hours, feed early morning and late evening. Use shade nets or plant trees around ponds to reduce direct sunlight. Monitor dissolved oxygen (DO) and apply aeration during hot afternoons.

Low Temperature: Reduce feeding rate and switch to highly digestible feed. Avoid handling fish during early mornings and nights. Provide windbreaks and increase pond depth to retain heat. Avoid use of antibiotics and strong chemicals during cold periods unless essential.

Litchi

Maintain sufficient moisture during hot summer of May to prevent fruit cracking. Regular watering is compulsory. Mulching with straw or grass around the tree basin is helpful for better soil moisture conservation. Fresh water spray upon the fruits and leaves during day time in hot and dry periods may also be helpful. To control fruit drop, spray NAA 4.5% W/W (Planofix) @ 0.2 ml/L at 10-12 days interval. For better fruit development, spray N-triacontanol (Miraculan) @ 0.5 ml/L at 15 days interval. Remove the insect affected dried branches and shoots arising from stem and branches during June. Spray Fipronil @ 1 ml/L to control leaf roller and leaf miner and Dicofol 18.5 EC @ 1.5 ml/L, two times at 7 days interval to prevent mites. To control stem borer/bark eating caterpillars, clean the holes from infested portion with help of cycle spoke and insert diesel or petrol-soaked cotton in the hole and plug the hole with wet clay soil. Spray dichlorvos (0.05%) to control the leaf eating weevil. Clean the orchard after last monsoon rain and paint the trunk up to 1-1.2 m by 1% Bordeaux mixture (lime: copper sulphate: water; 5:5:50). Prune and destroy infected shoots due to mite and spray Propergite 57 EC @ 2 ml/L to protect the plants from weevil and semilooper infestation.

Papaya

Seed must be soaked in water for overnight and treated with fungicide before sowing. Saplings are to be planted during September in raised beds to prevent water stagnation. After monsoon apply 500 g Ammonium Sulphate, 750 g SSP and 500 g MOP at 60 cm away of the basin of the established plants. Support papaya plants with stakes to prevent damage during strong winds. Remove weeds and apply mulch to conserve soil moisture. Spry Ethelyn 1 ml/4 L of water and Cycosol @ 2 ml/L of water at post kharif or early rabi season. Ensure proper drainage to avoid root rot and spray copper oxychloride @ 3 g/L to control fungal diseases. Use neem oil sprays to manage aphids and whiteflies in papaya. For sucking pest apply dimethoate 30 % @ 2 ml/L or imidacloprid 17.8 % @ 1 ml/3 L of water.

Banana

When the leaves become dry, cut them and dispose from the field to avoid insect. During the month of August, when the bunch becomes heavy, stake the plant with bamboo to avoid uprooting. After denavelling, apply GA3 1 g/30 L of water. To avoid Panama Wilt infestation, keep the orchard clean and remove the disease prone branches and spry Carbendazim 2 g/L of water, Streptocyclin 1 g/10 L of water. To avoid losses from scratching mite, use transparent polythene to cover the bunches. Apply Indoxacarb 14.5 % SC @ 1 ml/L or Spinosad 45 % SC @ 1 ml/L of water.

Coconut

Ensure proper drainage to avoid waterlogging. Apply 2-5 kg organic manure with 100 g urea, 150 g phosphate, and 75 g potash in a ring 25 cm wide and 1 m away from the trunk before the rainy season. Split doses of urea, SSP, MOP, boron, zinc and magnesium to be applied before (Apr.-May) and after monsoon (Aug.-Sept.) as per plant age. Use Metarhizium anisopliae in manure pits. Apply phorate 10 G + sand in crown and leaf axils at 6-month intervals for controlling

Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocynin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.